Python for practical data work

Bayram Uluad

Vektörel Bilişim × BTK

From first script to a reproducible report

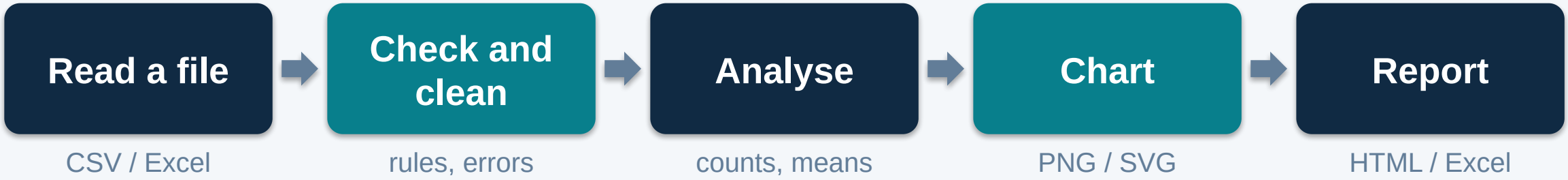
7 DAYS / LOCAL PYTHON / PRACTICAL DATA SKILLS

Who am I? Bayram Uluad

- 15 years in software development
- Web development, 3 years of Python
- Please speak one at a time
- Errors: send the file name and the last line

Official docs: docs.python.org/3/tutorial/index.html

Our destination



- Not a programming degree: confidence with your own data files
- Every step runs on your laptop; no cloud, no account
- Same story all week: telecom complaints and speed tests (synthetic data)

Official docs: docs.python.org/3/tutorial/index.html

Seven days, one story

Day	Topic	You will be able to...
1	Python basics	run a script, use types, read an error
2	Decisions and collections	filter and count records with if + for
3	Functions and files	read a CSV, write a clean CSV
4	Data model and SQLite	ask questions with SQL
5	Practical analysis	summarise like a PivotTable
6	Visualise and explain	choose and save a clear chart
7	A complete workflow	run one command, get a report

Official docs: docs.python.org/3/tutorial/index.html

How we learn

1. Predict the output, then run
2. Change one thing, run again
3. Explain the result to a neighbour
4. Ask with the file name and the last error line
5. Share files (.py / ZIP), not screenshots

Official docs: code.visualstudio.com/docs/python/python-tutorial

DAY 01

Day 1 • Think like a programmer

1 Welcome and first script

2 What is Python?

3 Values and types

4 Algorithms

5 Text and errors

6 Live coding and recap

TODAY'S GOAL

Open a .py file, run it, change a value, predict the result, read the error.

gun01/01_merhaba.py

Your first script

- A .py file is plain text
- print() shows a value in the terminal
- # starts a comment: not executed
- Save (Ctrl+S) before you run

```
# Script, comment and output.
# Python runs lines from top to bottom.
print("Merhaba BTK!")
```

OUTPUT

```
Merhaba BTK!
```

Official docs: docs.python.org/3/tutorial/introduction.html

`gun01/01_merhaba.py`

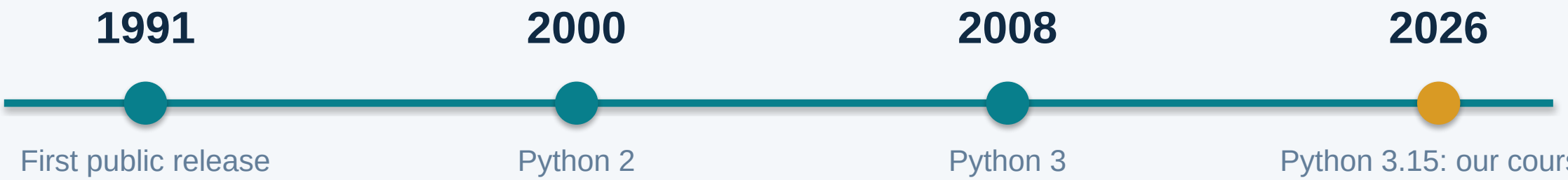
Two ways to run a script

1. Open the course folder: File → Open Folder
2. Open the .py file in the editor
3. Way 1: click the Run Python File button (top right)
4. Way 2: terminal → py gun01/01_merhaba.py
5. Read the output in the terminal panel

Official docs: docs.python.org/3/using/windows.html

gun01/05_python_ve_c.py

Python in one minute



- Created by Guido van Rossum
- Name comes from Monty Python, not the snake
- Readable, huge standard library, free
- Popular for automation and data work

Official docs: docs.python.org/3/faq/general.html#what-is-python

gun01/05_python_ve_c.py

Python and C: four differences

C

- Compile to .exe, then run
- Declare types: int x = 3;
- Braces { } make blocks
- Manage memory yourself
- int overflows at 2147483647

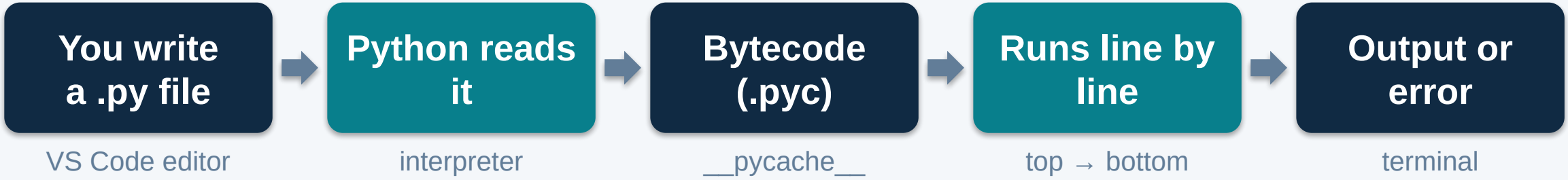
Python

- Run the .py with the interpreter
- Values carry the type
- Indentation makes blocks
- Lists grow automatically
- Big integers just work

Official docs: docs.python.org/3/tutorial/modules.html#compiled-python-files

gun01/06_script_nasil_calisir.py

How a script runs



- Lines run top to bottom, one at a time
- A later line can change a name; earlier output stays
- The first error stops the script

```
print("Adım 1: dosyanın en üstü")
il = "Ankara"
print("Adım 2: il değişkeni şu an:", il)
il = "İzmir"
print("Adım 3: il değişkeni artık:", il)
print("2026", "10", "14", sep="-")
```

Official docs: docs.python.org/3/tutorial/interpreter.html

gun01/07_degiskenler_ve_adlandirma.py

Variables: names for values

- name = value (= means assign)
- Letters, digits and _; never start with a digit
- il and Il are different names
- Names in ASCII, text in Turkish

```
KDV_ORANI = 0.20
paket_ucreti_tl = 1000
kdv = paket_ucreti_tl * KDV_ORANI
toplam_tutar = paket_ucreti_tl + kdv
print("Toplam tutar:", toplam_tutar, "TL")

il, operator_adi, sure = "Ankara", "Operatör A", 4
print("Çoklu atama:", il, operator_adi, sure)
```

OUTPUT

```
Toplam tutar: 1200.0 TL
```

Official docs: peps.python.org/pep-0008/

gun01/07_degiskenler_ve_adlandirma.py

A variable is not an Excel formula

- Excel recalculates a cell automatically
- Python calculates once, at that line
- Change the input → run the line again
- Order of lines matters

```
paket_ucreti_tl = 1000
kdv_tutari = paket_ucreti_tl * KDV_ORANI
paket_ucreti_tl = 2000
# kdv_tutari is STILL 200.0
print(kdv_tutari)
```

OUTPUT

```
200.0
```

Official docs: docs.python.org/3/tutorial/introduction.html

gun01/08_tur_donusumu.py

Five basic types

int	float	str	bool	None
10507	87.4	"İstanbul"	True	None
whole number: ID, count	decimal: speed, money	text, in quotes	yes / no	no value yet

- type(x) tells you the type
- None means 'no value yet' (an empty cell), not 0

```
sikayet_no = 10507
indirme_mbps = 87.4
il = "İstanbul"
print(type(il).__name__)
```

Official docs: docs.python.org/3/library/stdtypes.html

gun01/08_tur_donusumu.py

Text or number?

- "12" + "8" joins text → "128"
- int() and float() convert text to numbers
- Decimal comma breaks float()
- int() cuts, round() rounds

```
print("12" + "8")                # 128
print(int("12") + int("8"))    # 20
ham_deger = "12,5"
print(float(ham_deger.replace(",", ".")))
print(int(87.9), round(87.9))  # 87 88
```

OUTPUT

```
128
20
12.5
87 88
```

Official docs: docs.python.org/3/library/stdtypes.html

gun01/09_aritmetik_operatorler.py

Arithmetic operators

- / always gives a float: 10 / 2 = 5.0
- // whole part (Excel BÖLÜM)
- % remainder (Excel MOD)
- ** power (Excel ^)
- Brackets first, like Excel

```
print(17 / 5, 17 // 5, 17 % 5)   # 3.4 3 2
print(2 ** 10)                   # 1024
toplam_dakika = 150
saat = toplam_dakika // 60
dakika = toplam_dakika % 60
print(saat, "saat", dakika, "dakika")  # 2 30
```

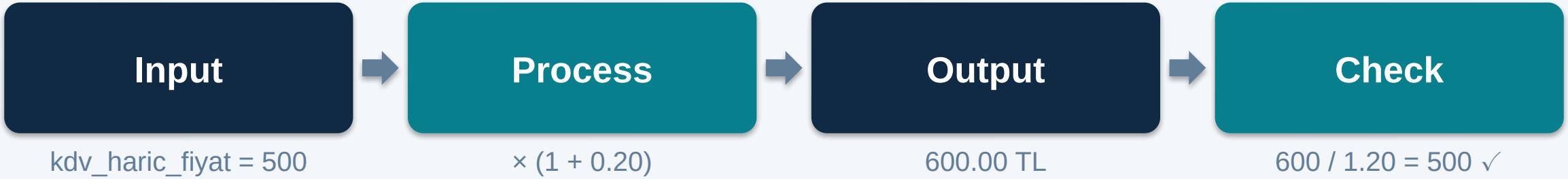
OUTPUT

```
3.4 3 2
1024
2 saat 30 dakika
```

Official docs: docs.python.org/3/reference/expressions.html#operator-precedence

gun01/11_algoritma_ornekleri.py

An algorithm is a clear recipe



- Write the steps in words before code
- Check the result with a small example by hand
- What if the list is empty? (Day 3)

```
KDV_ORANI = 0.20
kdv_haric_fiyat = 500                    # input
kdv_tutari = kdv_haric_fiyat * KDV_ORANI   # process
kdv_dahil_fiyat = kdv_haric_fiyat + kdv_tutari
print(f"KDV dahil fiyat: {kdv_dahil_fiyat:.2f} TL")
```

Official docs: docs.python.org/3/tutorial/introduction.html

gun01/11_algoritma_ornekleri.py

Mbps or MB/s?

- Operators sell megabits per second
- Download windows show megabytes per second
- 1 byte = 8 bits → divide by 8
- 100 Mbps ≈ 12.5 MB/s

```
paket_hizi_mbps = 100
dosya_boyutu_mb = 4500
hiz_mb_s = paket_hizi_mbps / 8
sure_sn = dosya_boyutu_mb / hiz_mb_s
print(f"{hiz_mb_s:.2f} MB/s")      # 12.50
print(f"{sure_sn:.0f} sn")         # 360
```

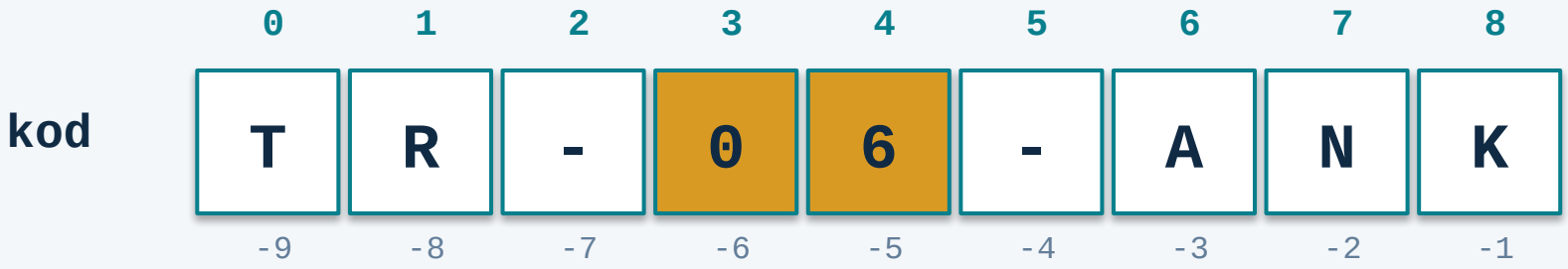
OUTPUT

```
12.50 MB/s
360 sn
```

Official docs: docs.python.org/3/tutorial/introduction.html

gun01/10_metin_islemleri.py

Text: indexing and slicing



- Index starts at 0
- [start:end] → end is NOT included
- Negative index counts from the end

```
kod = "TR-06-ANK"
print(len(kod))     # 9
print(kod[0:2])     # TR
print(kod[3:5])     # 06
print(kod[-3:])     # ANK
```

- Excel: SOLDAN, SAĞDAN, PARÇAAL, UZUNLUK

Official docs: docs.python.org/3/library/stdtypes.html#string-methods

gun01/10_metin_islemleri.py

Text methods and f-strings

- f-string: f"{il}: {adet}" (Excel &)
- :.2f two decimals, :.1% percent
- .strip() .upper() .replace() .split()
- Methods return a NEW text

```
ham_operator = "  operatör a  "
temiz = ham_operator.strip()
print(temiz.title())           # Operatör A
tarih = "2026-10-14"
parcalar = tarih.split("-")
print(f"Yıl: {parcalar[0]} | Ay: {parcalar[1]}")
print(f"Oran: {0.4567:.1%}")   # 45.7%
```

Official docs: docs.python.org/3/tutorial/inputoutput.html#formatted-string-literals

Excel → Python (Day 1)

Excel	Python
Name a cell: =tutar*kdv_orani	`kdv_tutari = tutar * KDV_ORANI`
'Number stored as text' warning	`"12" (str) vs 12 (int)`
SAYIYADÖNÜŞTÜR, DEĞER	`int(), float()`
BÖLÜM, MOD, ^	`//, %, **`
KIRP, BÜYÜKHARF, YERİNEKOY	`.strip(), .upper(), .replace()`
SOLDAN, SAĞDAN, PARÇAAL, UZUNLUK	`s[:3], s[-3:], s[2:5], len(s)`
Text to Columns	`.split(",")`
& joins text	`f"{il}: {adet}"`

Official docs: docs.python.org/3/tutorial/introduction.html

gun01/12_hata_mesaji_okuma.py

Read the error from the bottom

- Last line: error type + message
- Line above: the file and line number
- Fix one thing, run again
- Red text is help, not a punishment

```
Traceback (most recent call last):
  File "deneme.py", line 2, in <module>
    print(Il)
          ^^
NameError: name 'Il' is not defined.
Did you mean: 'il'?
```

Official docs: docs.python.org/3/tutorial/errors.html

`gun01/12_hata_mesaji_okuma.py`

Six common errors

Error	Typical cause	Quick fix
`SyntaxError`	print("Merhaba" (missing ")")	Check ( ) quotes and :
`IndentationError`	if block not indented	4 spaces inside the block
`NameError`	birim_fiyat not defined / typo	Define before use; check spelling
`TypeError`	"5" + 3	int("5") + 3 or "5" + str(3)
`ValueError`	float("87,4")	replace(",", ".") first
`FileNotFoundError`	wrong folder or file name	Run from the course folder

Official docs: docs.python.org/3/tutorial/errors.html

gun01/13_hiz_raporu_bos.py

Live coding • a speed report

TOGETHER Open 13_hiz_raporu_bos.py and fill TODO 1-7

CHECK Run after every TODO; compare with 13_hiz_raporu.py

CHANGE Use measurement 1: 1000 Mbps package, 53.6 Mbps measured

EXTRA Add ping and upload speed (alanlar[7], alanlar[8])

```
py gun01/13_hiz_raporu_bos.py
py gun01/13_hiz_raporu.py
```

```
satir = ("3,2026-08-23,Ankara,Operatör B,"
         "Fiber,100,77.7,36.0,10.6")
alanlar = satir.split(",")
il = alanlar[2]
paket_hizi_mbps = int(alanlar[5])
indirme_mbps = float(alanlar[6])
```

OUTPUT

```
TODO sayısı: 7
1000 MB indirme süresi: 1 dk 42 sn
```

Official docs: docs.python.org/3/tutorial/introduction.html

If it does not run (Windows)

- python opens the Microsoft Store → use py
- 'py' is not recognized → try python or the Run button
- No such file → open the course folder, not one file
- Strange letters (Ã§) → save as UTF-8 (bottom right)
- File names: ASCII, no spaces → calisma1.py

Official docs: docs.python.org/3/using/windows.html

Exit ticket • Day 1

1

What is the difference between saving and running?

2

What do 17 // 5 and 17 % 5 give?

3

Why are "5" and 5 different?

4

Where do you start reading an error?

Official docs: docs.python.org/3/tutorial/index.html

DAY 02

Day 2 • Decisions and collections

1 Comparisons and logic

2 if / elif / else

3 Lists and for loops

4 Counters and dictionaries

5 Tuples, sets, shortcuts

6 Mini project

TODAY'S GOAL

Filter and count a small complaint table with for + if; look values up with a dictionary.

gun02/05_karsilastirma_mantiksal.py

A comparison answers True or False

- Like =B2>10 in an Excel cell
- == asks, = assigns
- "Fatura " is not "Fatura" (space!)
- Text comparison is case-sensitive

```
sure = 12
hedef = 10
print(sure > hedef)                 # True
print(sure == hedef)                # False
hedef_asildi = sure > hedef
print("Fatura " == "Fatura")        # False
print(10 == "10")                   # False
```

Official docs: docs.python.org/3/library/stdtypes.html#comparisons

`gun02/05_karsilastirma_mantiksal.py`

and / or / not

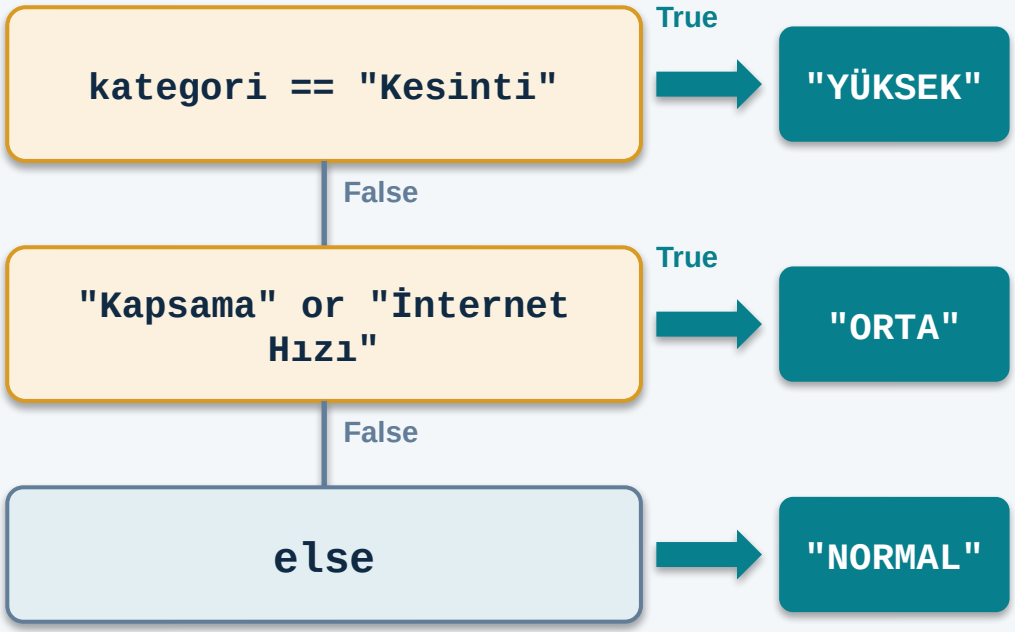
cozuldu	sure > hedef	and	or
True	True	True	True
True	False	False	True
False	True	False	True
False	False	False	False

Excel: VE = and, YADA = or, DEĞİL = not. not True → False

Official docs: docs.python.org/3/library/stdtypes.html

gun02/06_sikayet_onceliklendirme.py

if / elif / else: first match wins



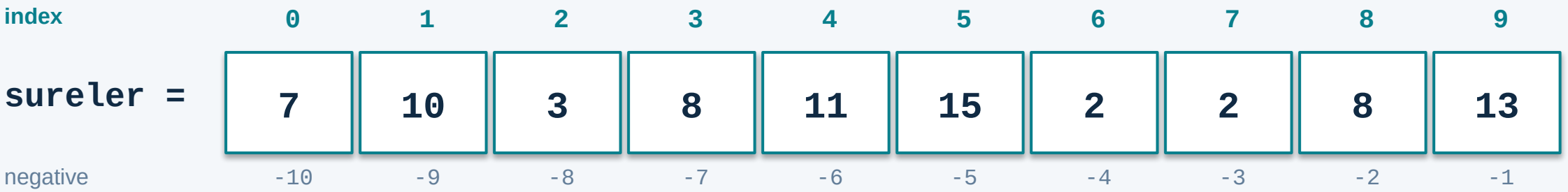

```
kategori = "Kesinti"
if kategori == "Kesinti":
    oncelik = "YÜKSEK"
elif kategori in ("Kapsama", "İnternet Hızı"):
    oncelik = "ORTA"
else:
    oncelik = "NORMAL"
print(oncelik)
```

- Colon, then a 4-space block
- Only the first true branch runs
- else catches the rest

Official docs: docs.python.org/3/tutorial/controlflow.html

gun02/07_listeler.py

A list is one Excel column



- len, sum, min, max = BAĞ_DEĞ_SAY, TOPLA, MİN, MAK
- Index starts at 0, the last is [-1]
- sorted() gives a new list; .sort() returns None

```
sureler = [7, 10, 3, 8, 11, 15, 2, 2, 8, 13]
adet = len(sureler)
toplam = sum(sureler)
print(f"Adet: {adet} | Toplam: {toplam}")
print(sureler[0], sureler[-1], sureler[2:5])
```

OUTPUT

```
Adet: 10 | Toplam: 79
7 13 [3, 8, 11]
```

Official docs: docs.python.org/3/tutorial/datastructures.html

gun02/08_for_range_enumerate_zip.py

for: drag the formula down

- The indented block runs once per item
- The line after the loop runs once
- Name the loop variable well: for il in iller
- range(1, 6) → 1..5 (end not included)

```
iller = ["İstanbul", "Ankara", "Antalya"]
for il in iller:
    print("İl:", il)       # inside: repeats
print("Döngü bitti.")      # outside: once

for ay in range(1, 4):
    print(f"2026-{ay:02d} raporu")
```

Official docs: docs.python.org/3/tutorial/controlflow.html

gun02/08_for_range_enumerate_zip.py

enumerate and zip

- enumerate → a 'row number' column
- zip → read A2 and B2 together
- zip stops at the shorter list (silently!)
- for + if = filter

```
iller = ["İstanbul", "Ankara", "Antalya"]
adetler = [5, 3, 2]
for sira, il in enumerate(iller, start=1):
    print(f"{sira}. {il}")
for il, adet in zip(iller, adetler):
    if adet >= 3:
        print(f"Yoğun il: {il} ({adet})")
```

Official docs: docs.python.org/3/library/functions.html#enumerate

gun02/09_biriktirme_sayac.py

Start • update • use: the accumulator

- 1 start before the loop
- 2 update inside the loop
- 3 use after the loop
- Excel: TOPLA, EĞERSAY, ETOPLA

```
sureler = [7, 10, 3, 8, 11, 15, 2, 2, 8, 13]
toplam = 0                    # 1 start
sayac = 0
for sure in sureler:
    toplam += sure            # 2 update
    if sure > 10:
        sayac += 1
print(toplam, sayac)          # 3 use: 79 3
```

Official docs: docs.python.org/3/tutorial/controlflow.html

gun02/10_sozlukler.py

A dictionary is a lookup table (DÜŞEYARA)




```
hedefler = {"Fatura": 7, "İnternet Hızı": 10,
            "Kesinti": 5, "Kapsama": 20}
print("Kesinti hedefi:", hedefler["Kesinti"], "gün")
print(hedefler.get("Roaming", 10))   # default
```

OUTPUT

```
Kesinti hedefi: 5 gün
10
```

- key → value; keys are unique
- hedefler[kategori] = DÜŞEYARA
- .get(key, default) = EĞERHATA

Official docs: docs.python.org/3/tutorial/datastructures.html#dictionaries

gun02/10_sozlukler.py

Count with a dictionary (PivotTable 'Count')

- One key per city, value = count
- .get(il, 0): start from 0 if new
- Same idea as Excel COUNTIF per row
- Day 3: the same code on 600 rows

```
iller = ["İstanbul", "Ankara", "İstanbul",
         "Konya", "İstanbul", "Ankara"]
sayim = {}
for il in iller:
    sayim[il] = sayim.get(il, 0) + 1
print(sayim)
```

OUTPUT

```
{'İstanbul': 3, 'Ankara': 2, 'Konya': 1}
```

Official docs: docs.python.org/3/tutorial/datastructures.html#dictionaries

gun02/11_tuple_set.py

Choose a collection

Type	Looks like	Use it for	Excel idea
`list`	`[7, 10, 3]`	ordered values, duplicates OK	a column
`dict`	`{"Fatura": 7}`	look up by name	DÜŞEYARA table
`tuple`	`(39.93, 32.86)`	fixed pair / record	a locked row
`set`	`{"Ankara", "İzmir"}`	unique values	Remove Duplicates

```
iller = ["İstanbul", "Ankara", "İstanbul", "İzmir"]
print(sorted(set(iller)))      # unique
print(len(set(iller)))         # 3
```

Official docs: docs.python.org/3/tutorial/datastructures.html

gun02/12_pythona_ozgu_yapilar.py

Python shortcuts you will meet

- List comprehension = for + if + append
- Read it: 'each s, if s > 10, keep s'
- One-line if = Excel EĞER
- Goal: read them in ready scripts

```
sureler = [7, 10, 3, 8, 11, 15, 2, 2, 8, 13]
uzunlar = []
for s in sureler:
    if s > 10:
        uzunlar.append(s)
uzunlar_kisa = [s for s in sureler if s > 10]
print(uzunlar_kisa)                # [11, 15, 13]
etiket = "Gecikmeli" if 13 > 10 else "Zamanında"
```

Official docs: docs.python.org/3/tutorial/datastructures.html#list-comprehensions

gun02/13_while_break_continue.py

while: repeat while a condition holds

- Use when you do not know how many rounds
- Always add a safety limit
- break: stop now • continue: skip this one
- Stuck? Click the terminal, press Ctrl+C

```
aylik_sikayet = [30, 35, 47, 52, 41, 38]
birikimli = 0
ay = 0
while birikimli <= 100 and ay < len(aylik_sikayet):
    birikimli += aylik_sikayet[ay]
    ay += 1     # without this: infinite loop!
print(ay, birikimli)          # 3 112
```

Official docs: docs.python.org/3/tutorial/controlflow.html

gun02/14_tablo_liste_sozluk.py

A table is a list of dictionaries

#	no	il	kategori	durum	sure
0	10507	İstanbul	Kesinti	Reddedildi	7
1	10527	Ankara	Fatura	Çözüldü	10
2	10550	Antalya	İnternet Hızı	İşlemde	None
3	10045	Konya	Fatura	Çözüldü	3

- Each row = one dictionary
- The list holds all rows
- tablo[1]["il"] → row 1, column il
- Day 3: a CSV file is read exactly like this

```
print(tablo[1]["il"])           # Ankara
ankara = [s for s in tablo if s["il"] == "Ankara"]
print("Ankara satırları:", len(ankara))
sureler = [s["sure"] for s in tablo
           if s["sure"] is not None]
```

Official docs: docs.python.org/3/tutorial/datastructures.html

Excel → Python (Day 2)

Excel	Python
=B2>10 → DOĞRU/YANLIŞ	`sure > 10 → True/False`
VE, YADA, DEĞİL	`and, or, not`
Nested EĞER / ÇOKEĞER	`if / elif / else`
EĞERSAY, ETOPLA	`counter: for + if + += 1`
Drag the formula down	`for loop`
AutoFilter / FİLTRE()	`[s for s in tablo if s["il"] == "Ankara"]`
DÜŞEYARA / EĞERHATA	`hedefler[kategori] / .get(k, 10)`
PivotTable 'Count'	`sayim[il] = sayim.get(il, 0) + 1`
Remove Duplicates	`set(iller)`

Official docs: docs.python.org/3/tutorial/datastructures.html

gun02/15_mini_proje_bos.py

Mini project • complaint summary

CORE Count complaints per city (TODO 1-2)

CORE Find the longest resolution

EXTRA How many exceeded the target? Which denominator?

EXTRA Add a count per operator

```
py gun02/15_mini_proje_bos.py
py gun02/15_mini_proje.py
```

OUTPUT

```
İskelet çalıştı. TODO sayısı: 5
Hedef aşan: 5 / 12 sonuçlanmış şikâyet (%41.7)
```

Official docs: docs.python.org/3/tutorial/datastructures.html

Exit ticket • Day 2

1

What is the difference between = and ==?

2

Which lines does a for loop repeat?

3

How do you look up a target day by category?

4

In 5 / 12, why 12 and not 15?

Official docs: docs.python.org/3/tutorial/index.html

DAY 03

Day 3 • Reusable scripts and files

1 Functions

2 Modules and dates

3 Files and CSV

4 Write CSV and JSON

5 Bad rows and parameters

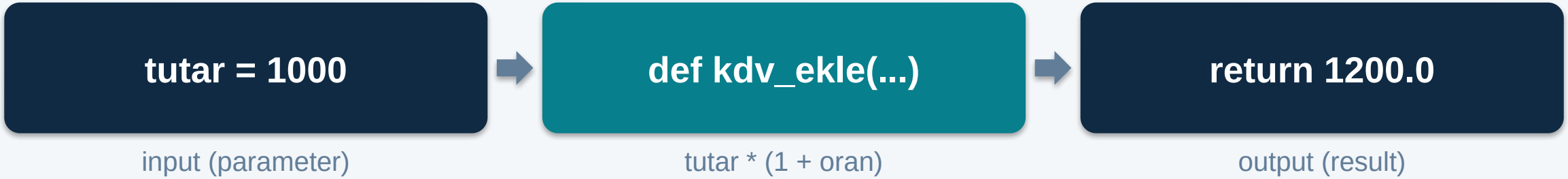
6 Mini SLA report

TODAY'S GOAL

Write your own formula (function), read a 600-row CSV, write a clean file for Excel.

gun03/05_fonksiyon_parametreleri.py

A function is a small machine



- def defines it; calling it runs it
- Parameters come in, return gives a value back
- oran=0.20 is a default value
- Name the argument when it helps: oran=0.10

```
def kdv_ekle(tutar, oran=0.20):
    sonuc = tutar * (1 + oran)
    return round(sonuc, 2)

print(kdv_ekle(1000))              # 1200.0
print(kdv_ekle(100, oran=0.10))    # 110.0
```

Official docs: docs.python.org/3/tutorial/controlflow.html#defining-functions

gun03/05_fonksiyon_parametreleri.py

return gives back; print only shows

- print → you see it, you cannot reuse it
- No return → the function gives None
- return can give two values: net, kdv
- Keep calculation and display apart

```
def kdv_yazdir(tutar):
    print(round(tutar * 1.2, 2))

deger = kdv_yazdir(100)     # shows 120.0
print(deger)                # None!

def kdv_ayir(kdv_dahil_tutar, oran=0.20):
    net = round(kdv_dahil_tutar / (1 + oran), 2)
    return net, round(kdv_dahil_tutar - net, 2)
```

Official docs: docs.python.org/3/tutorial/controlflow.html#defining-functions

gun03/06_hazir_moduller.py

import: ready-made tools

- Standard library comes with Python
- No internet, no install, no IT ticket
- statistics = ORTALAMA, ORTANCA, STDSAPMA.S
- random.seed(42) → same 'random' result

```
import math
import statistics
import random

print("Gereken sayfa:", math.ceil(610 / 50))
print(statistics.median([7, 10, 3, 8, 11]))
random.seed(42)
print(random.sample(range(10001, 10601), 3))
```

Official docs: docs.python.org/3/tutorial/stdlib.html

gun03/07_kendi_modulum.py

Your own module: yardimcilar.py



- A module is just a .py file with functions
- import yardimcilar → use yardimcilar.kdv_ekle()
- if __name__ == "__main__": runs only when the file runs directly

```
import yardimcilar
from yardimcilar import ortalama

print("KDV dahil:", yardimcilar.kdv_ekle(1000))
kayitlar = yardimcilar.csv_oku("sikayetler.csv")
print("Şikâyet sayısı:", len(kayitlar))
```

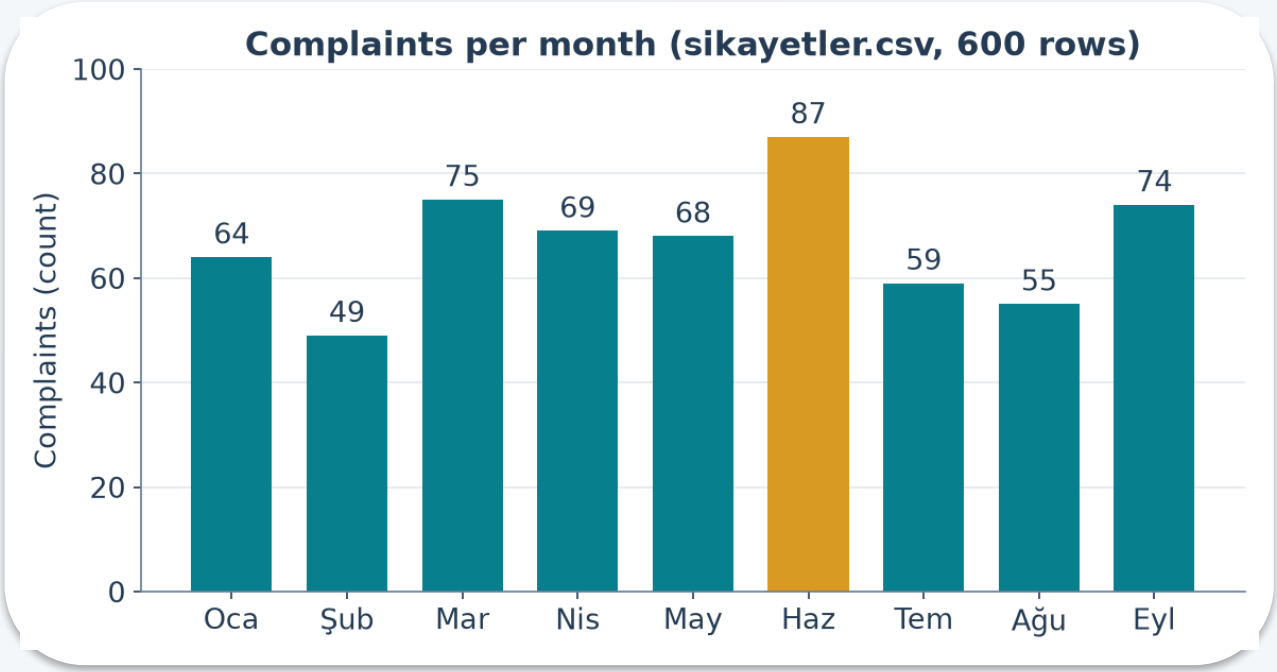
Official docs: docs.python.org/3/tutorial/modules.html

gun03/08_tarihler.py

Dates: difference, add days, month name

- date.fromisoformat("2026-03-02")
- (kapanis - basvuru).days = Excel =B2-A2
- + timedelta(days=10) = Excel =A2+10
- strptime/strftime: 02.03.2026 ↔ date

```
from datetime import date, timedelta
basvuru = date(2026, 3, 2)
kapanis = date(2026, 3, 16)
print((kapanis - basvuru).days)  # 14
print(basvuru + timedelta(days=10))
```



Official docs: docs.python.org/3/library/datetime.html

gun03/09_log_dosyasi.py

Open, write, append, read a file

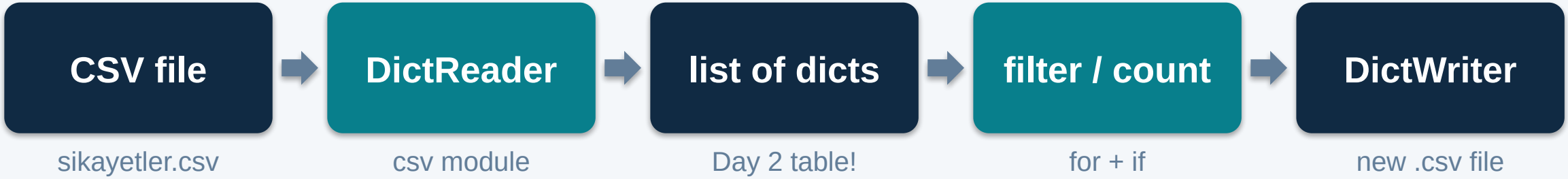
- with … closes the file for you
- Always say encoding="utf-8"
- "w" overwrites • "a" appends • "r" reads
- Build paths from the script: ROOT / "outputs"

```
ROOT = Path(__file__).resolve().parents[1]
cikti = ROOT / "outputs" / "gun03"
log_dosyasi = cikti / "islem_log.txt"
with open(log_dosyasi, "a", encoding="utf-8") as f:
    f.write("2026-10-14 09:03 BILGI Rapor bitti\n")
with open(log_dosyasi, encoding="utf-8") as f:
    for satir in f:
        print(satir.rstrip())
```

Official docs: docs.python.org/3/tutorial/inputoutput.html#reading-and-writing-files

gun03/10_sikayet_csv_okuma.py

CSV in, CSV out



- Everything read from CSV is TEXT: '7' is not 7
- Empty cell is "" → not 0
- utf-8-sig + ; → opens correctly in Turkish Excel

```
with (ROOT / "data" / "sikayetler.csv").open(
        encoding="utf-8", newline="") as dosya:
    kayitlar = list(csv.DictReader(dosya))
print("Toplam şikâyet:", len(kayitlar))
print(kayitlar[0]["il"])
```

Official docs: docs.python.org/3/library/csv.html

gun03/10_sikayet_csv_okuma.py

An empty cell is not zero

- İşlemde rows have no resolution time
- Skip them with continue
- Convert text to int only after the check
- Excel: ORTALAMAEĞER skips blanks too

```
cozulen_sureler = []
for kayit in kayitlar:
    metin = kayit["cozum_suresi_gun"]
    if metin == "":
        continue                    # skip blank
    if kayit["durum"] == "Çözüldü":
        cozulen_sureler.append(int(metin))
toplam = sum(cozulen_sureler)
ortalama = toplam / len(cozulen_sureler)
print(f"{ortalama:.2f} gün")      # 9.60
```

Official docs: docs.python.org/3/library/csv.html

gun03/11_csv_filtrele_yaz.py

Filter and write a CSV for Excel

- Filter = Excel AutoFilter
- DictWriter writes header + rows
- newline="" avoids blank lines on Windows
- delimiter=";" for Turkish Excel

```
secilenler = [k for k in kayitlar
              if k["il"] == "Ankara"
              and k["kategori"] == "İnternet Hızı"]
with open(dosya1, "w", encoding="utf-8-sig",
          newline="") as f:
    alanlar = list(kayitlar[0])
    yazici = csv.DictWriter(f, fieldnames=alanlar)
    yazici.writeheader()
    yazici.writerows(secilenler)
```

Official docs: docs.python.org/3/library/csv.html

gun03/12_json_okuma_yazma.py

JSON: save a summary with structure

- JSON object ↔ Python dict, array ↔ list
- json.dump writes, json.load reads
- ensure_ascii=False keeps İ, ş, ğ readable
- Good for settings and summaries

```
import json
rapor = {"kaynak": "data/sikayetler.csv",
         "toplam_kayit": len(kayitlar),
         "iller": ozet}
with open(yol, "w", encoding="utf-8") as f:
    json.dump(rapor, f, ensure_ascii=False,
              indent=2)
with open(yol, encoding="utf-8") as f:
    geri = json.load(f)
```

Official docs: docs.python.org/3/library/json.html

gun03/13_hatali_satirlar.py

Bad rows: catch, record, continue

- try: the risky step
- except ValueError: what to do on that error
- else: runs if no error • finally: always
- Never a bare except:

```
for no, kayit in enumerate(ham, start=2):
    try:
        temiz = satiri_dogrula(kayit)
    except ValueError as hata:
        hatalar.append(f"Satır {no}: {hata}")
    else:
        temizler.append(temiz)
print("Temiz:", len(temizler))
print("Hatalı:", len(hatalar))
```

Official docs: docs.python.org/3/tutorial/errors.html

gun03/14_parametreli_script.py

A script that takes parameters

- sys.argv = words typed after the file name
- No parameter → a default value
- Invalid value → clear message + sys.exit(1)
- The Run button sends no parameters

```
import sys
VARSAYILAN_IL = "Ankara"
if len(sys.argv) > 1:
    il = sys.argv[1].strip()
else:
    il = VARSAYILAN_IL
secilen = [k for k in kayitlar if k["il"] == il]
print(f"{il}: {len(secilen)} şikâyet")
```

Official docs: docs.python.org/3/library/sys.html#sys.argv

Excel → Python (Day 3)

Excel	Python
Your own formula	`def kdv_ekle(tutar, oran=0.20):`
Add-in / function library	`import statistics, import yardimcilar`
=B2-A2 (days between)	`(kapanis - basvuru).days`
=A2+10	`basvuru + timedelta(days=10)`
Save As 'CSV UTF-8'	`encoding="utf-8-sig"`
AutoFilter, then copy	`for + if, then csv.DictWriter`
ORTALAMAEĞER	`skip blanks, then sum / len`
EĞERHATA	`try / except ValueError`

Official docs: docs.python.org/3/tutorial/stdlib.html

Windows file traps

You see	Why	Fix
`PermissionError [Errno 13]`	The output is open in Excel	Close Excel, run again
`Ã¼ / Ä° letters`	Wrong encoding	encoding="utf-8-sig" for Excel
`CSV opens in one column`	Turkish Excel expects ;	Use the _excel_tr.csv file
`Blank line between rows`	newline="" missing	open(..., newline="")
`"C:\Users\..." error`	\U is an escape	Use Path or r"C:\..."

Official docs: docs.python.org/3/tutorial/inputoutput.html#reading-and-writing-files

gun03/15_mini_sla_raporu_bos.py

Mini project • SLA report by category

TOGETHER Fill TODO 1-6 in 15_mini_sla_raporu_bos.py

CHECK Run after every TODO; counters change

OPEN outputs/gun03/sla_raporu.csv in Excel

DISCUSS İnternet Hızı 61% late: target or process?

```
py gun03/15_mini_sla_raporu_bos.py
py gun03/15_mini_sla_raporu.py
```

OUTPUT

```
İskelet çalıştı: 600 kayıt okundu
Toplam çözülen: 415
```

Official docs: docs.python.org/3/library/csv.html

Exit ticket • Day 3

1

What is the difference between return and print?

2

Why is an empty cell not 0?

3

Which encoding for a CSV that goes to Excel?

4

What does PermissionError usually mean?

Official docs: docs.python.org/3/tutorial/index.html

DAY 04

Day 4 • Data models and SQLite

1 Data model and classes

2 Load SQLite, SELECT

3 GROUP BY and HAVING

4 JOIN, safe writes, ?

5 Excel → database → CSV

6 Mini project, Day 5 check

TODAY'S GOAL

Describe data as tables and keys, then ask questions with SQL — no server needed.

`gun04/05_veri_modeli_sikayet.py`

Excel words, database words

Excel	Database	Our example
Workbook file	Database file	`sikayet.sqlite`
Sheet	Table	`sikayetler`
Row	Record	`one complaint`
Column header	Field	`il, kategori, durum`
Unique ID column	Primary key (PK)	`sikayet_no`
Column used by DÜŞEYARA	Foreign key (FK)	`kategori → hedefler`
Empty cell	NULL	`cozum_suresi_gun`

Official docs: sqlite.org/datatype3.html

gun04/05_veri_modeli_sikayet.py

Our data model



- One city has many complaints (1 → N)
- Each complaint category has one target
- The key makes the link explicit

Official docs: sqlite.org/lang_createtable.html

gun04/06_sikayet_sinifi.py

A class is a form template

- class = the empty form
- object = one filled-in form
- self = 'this form'
- Recognise it in ready code; a dict is often enough

```
class Sikayet:
    def __init__(self, sikayet_no, il,
                 kategori, durum, sure=None):
        self.sikayet_no = sikayet_no
        self.il = il
        self.durum = durum
        self.sure = sure

    def cozuldu_mu(self):
        return self.durum == "Çözüldü"
```

Official docs: docs.python.org/3/tutorial/classes.html

gun04/06_sikayet_sinifi.py

Create and use one object

- Calling the class runs __init__
- object.attribute reads a field
- object.method() runs a behaviour

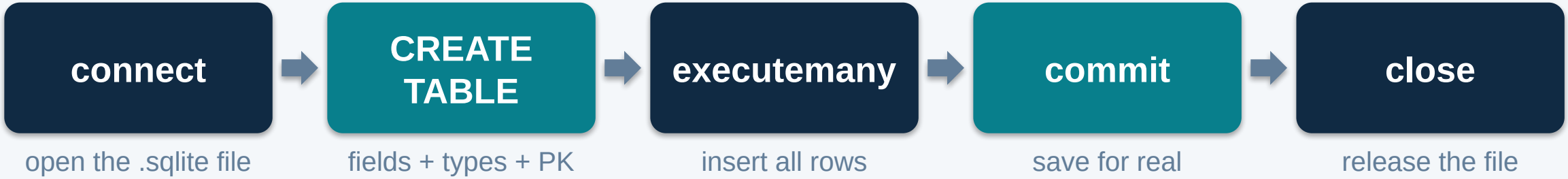
- Each object keeps its own data

```
s = Sikayet(10527, "Ankara", "Fatura",
            "Çözüldü", 10)
print(s.il)                  # Ankara
print(s.cozuldu_mu())        # True
print(s.sla_asildi_mi(7))    # True: 10 > 7
```

Official docs: docs.python.org/3/tutorial/classes.html

gun04/07_sqlite_yukleme.py

Load a CSV into SQLite



- SQLite = one file, no server, no install
- sqlite3 is in the standard library
- Empty text → None → NULL in the database

```
baglanti = sqlite3.connect(db_yolu)
baglanti.executemany(
    "INSERT INTO sikayetler VALUES "
    "(?,?,?,?,?,?,?,?,?,?,?)",
    satirlar)
baglanti.commit()
baglanti.close()
```

Official docs: docs.python.org/3/library/sqlite3.html

gun04/08_select_where_order.py

SELECT … WHERE … ORDER BY

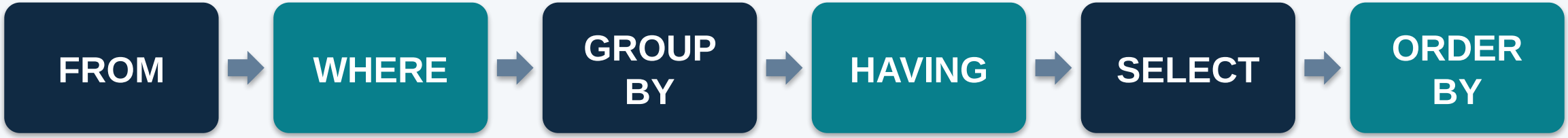
- SELECT: which columns
- FROM: which table (sheet)
- WHERE: which rows (AutoFilter)
- ORDER BY … DESC, LIMIT 5: sort, top 5
- Text values in single quotes

```
SELECT sikayet_no, tarih, operator,
       cozum_suresi_gun
FROM sikayetler
WHERE il = 'Ankara'
  AND kategori = 'Fatura'
  AND durum = 'Çözüldü'
ORDER BY cozum_suresi_gun DESC;
```

Official docs: sqlite.org/lang_select.html

`gun04/09_toplama_gruplama.py`

The order SQL works in



- WHERE filters rows, HAVING filters groups
- Every plain column in SELECT must be in GROUP BY
- Read the query in this order when it surprises you

Official docs: sqlite.org/lang_select.html

gun04/09_toplama_gruplama.py

GROUP BY = PivotTable

- COUNT(*) counts rows; COUNT(col) skips NULL
- AVG ignores NULL
- AS gives a column a clear name
- HAVING = PivotTable value filter

```
SELECT operator,
       COUNT(*) AS sikayet,
       ROUND(AVG(cozum_suresi_gun), 2) AS ort_sure
FROM sikayetler
GROUP BY operator
ORDER BY sikayet DESC;

SELECT il, COUNT(*) FROM sikayetler
GROUP BY il HAVING COUNT(*) > 100;
```

Official docs: sqlite.org/lang_aggfunc.html

gun04/10_join_hedefler.py

JOIN = DÜŞEYARA

- ON says which column matches which
- INNER JOIN keeps matches only
- LEFT JOIN keeps all rows (#YOK → NULL)
- Check: row count must not change

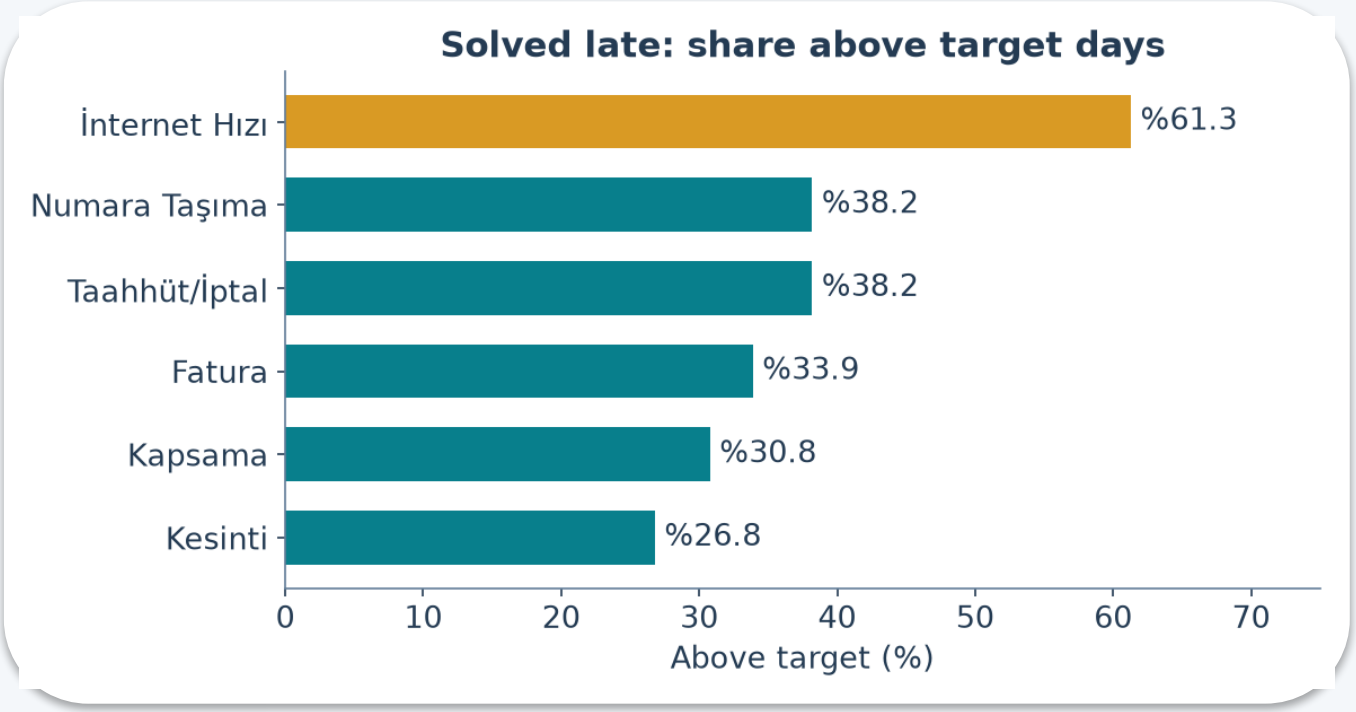
```
SELECT s.kategori, h.hedef_gun,
       COUNT(*) AS cozulen,
       SUM(s.cozum_suresi_gun > h.hedef_gun) AS asan
FROM sikayetler AS s
JOIN hedefler AS h ON s.kategori = h.kategori
WHERE s.durum = 'Çözüldü'
GROUP BY s.kategori, h.hedef_gun;
```

Official docs: sqlite.org/lang_select.html

gun04/10_join_hedefler.py

Same answer, two tools

- Day 3: Python loops → 61.3%
- Day 4: SQL JOIN + GROUP BY → 61.3%
- Two routes agree → we can trust the number
- Late = solved AND days > target



Official docs: sqlite.org/lang_aggfunc.html

gun04/11_update_delete_guvenli.py

Change data safely

- Work on a copy first
- SELECT COUNT(*) before UPDATE/DELETE
- Compare with rowcount
- commit if expected, otherwise rollback

```
kosul = "itiraz_tutari_tl IS NOT NULL"
once = baglanti.execute(
    f"SELECT COUNT(*) FROM sikayetler WHERE {kosul}"
).fetchone()[0]
imlec = baglanti.execute(
    "UPDATE sikayetler SET kdv_dahil_tl = "
    "ROUND(itiraz_tutari_tl * 1.20, 2) "
    f"WHERE {kosul}")
if imlec.rowcount == once:
    baglanti.commit()
else:
    baglanti.rollback()
```

Official docs: sqlite.org/lang_update.html

gun04/12_sql_injection_parametre.py

Use ? for values, never f-strings

- ? keeps data separate from the SQL text
- A quote in the data cannot break the query
- Bad input cannot leak all rows
- One value still needs a tuple: (il,)

```
kotu_girdi = "x' OR '1'='1"
sql = ("SELECT COUNT(*) FROM sikayetler "
       f"WHERE il = '{kotu_girdi}'")
print(baglanti.execute(sql).fetchone()[0])  # 600!

print(baglanti.execute(
    "SELECT COUNT(*) FROM sikayetler WHERE il = ?",
    (kotu_girdi,)).fetchone()[0])           # 0
```

Official docs: docs.python.org/3/library/sqlite3.html#sqlite3-placeholders

gun04/13_excel_stdlib_okuma.py

An .xlsx file is a ZIP of XML files



- No package? Python can still read Excel
- openpyxl does the same with less code (optional)
- Dates may arrive as serial numbers (46023)

```
xlsx = dy.DATA / "btk_ornek_veri.xlsx"
with zipfile.ZipFile(xlsx) as arsiv:
    print(len(arsiv.namelist()))
hedefler = dy.xlsx_sayfa_oku(xlsx, "Hedefler")
print("Hedefler:", len(hedefler), "kategori")
```

Official docs: docs.python.org/3/library/zipfile.html

gun04/15_sorgu_csv_rapor.py

Query result → CSV for Excel

- cursor.description gives the column names
- fetchall() gives the rows as tuples
- csv.writer (not DictWriter) for tuples
- Read it back: did we write 600?

```
basliklar = [s[0] for s in imlec.description]
satirlar = imlec.fetchall()
with open(dosya, "w", encoding="utf-8-sig",
          newline="") as f:
    yazici = csv.writer(f)
    yazici.writerow(basliklar)
    yazici.writerows(satirlar)
```

Official docs: docs.python.org/3/library/csv.html

Excel → SQL (Day 4)

Excel	SQL
Hide / show columns	`SELECT col1, col2`
AutoFilter	`WHERE il = 'Ankara'`
Sort Z → A	`ORDER BY sure DESC`
PivotTable rows + Count	`GROUP BY il … COUNT(*)`
PivotTable value filter	`HAVING COUNT(*) > 100`
DÜŞEYARA	`JOIN hedefler ON …`
Rows where DÜŞEYARA gives #YOK	`LEFT JOIN … IS NULL`
Ctrl+Z	`rollback() (before commit only)`

Official docs: sqlite.org/lang_select.html

gun04/16_mini_proje_bos.py

Mini project • three questions in SQL

CORE S1: which city has the lowest solve rate?

EXTRA S2 (JOIN): which category is most often late?

EXTRA S3: which operator delivers the lowest speed ratio?

DISCUSS Is a 2-point difference a real difference?

```
py gun04/16_mini_proje_bos.py
py gun04/16_mini_proje.py
```

OUTPUT

```
Rapor: outputs/gun04/mini_proje_rapor.csv (3 soru)
```

Official docs: sqlite.org/lang_select.html

Exit ticket • Day 4

1. What does one row of sikayetler represent?

2. WHERE or HAVING: which filters groups?

3. Why must the row count stay 600 after JOIN?

4. Why do we use ? instead of an f-string?

```
py -c "import pandas, numpy, openpyxl; print('PAKET TAMAM')"
```

Official docs: docs.python.org/3/tutorial/index.html

DAY 05

Day 5 • Practical analysis

1 Analysis thinking

2 Lists vs NumPy

3 pandas first look

4 Group, pivot, merge, dates

5 Missing and dirty data, Excel

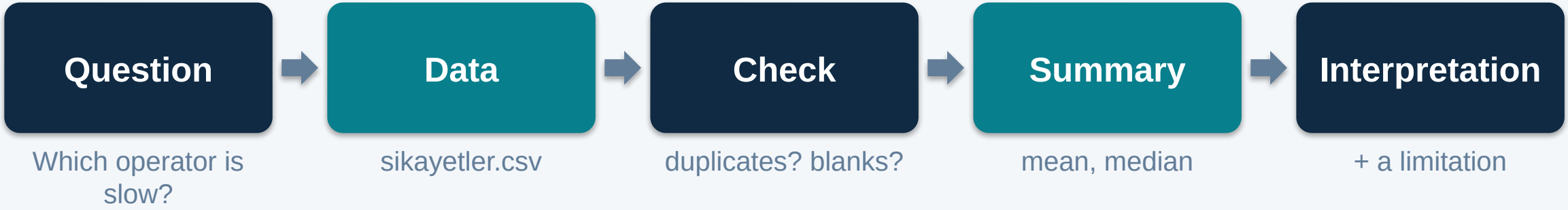
6 Operator scorecard

TODAY'S GOAL

Do with Python what you do with PivotTable, Filter and DÜŞEYARA — and put the result back in Excel.

gun05/07_analiz_dusuncesi.py

Analysis is more than a number



- Write the question first
- Check the data before you summarise
- Report count, measure and unit

```
numaralar = [k["sikayet_no"] for k in kayitlar]
assert len(set(numaralar)) == len(numaralar), \
    "Tekrar eden şikâyet numarası var!"
sureler = [int(k["cozum_suresi_gun"])
           for k in kayitlar
           if k["durum"] == "Çözüldü"]
print("Çözülen şikâyet:", len(sureler))
```

Official docs: docs.python.org/3/library/statistics.html

Two routes, same answers

Topic	A: with packages	B: standard library only
Descriptive statistics	07	07 (statistics)
Many values at once	09 NumPy	08 lists
First look, filter, sort	10 pandas	12 (parts 1-4)
Group, pivot, DÜŞEYARA	11 pandas	12 (parts 5-8)
Missing and dirty data	13, 15	14
Excel read / write	04, 16	17 + xlsx_mini.py
Operator scorecard	18 → 20	18 → 19

Packages change the tool, not the evidence

Official docs: pandas.pydata.org/docs/getting_started/intro_tutorials/index.html

gun05/07_analiz_dusuncesi.py

Describe a column in five numbers

- mean = ORTALAMA, median = ORTANCA
- mode = ENÇOK_OLAN
- stdev = STDSAPMA.S (n−1)
- quantiles = DÖRTTEBİRLİK.DHL (inclusive)

```
import statistics
ortalama = statistics.mean(sureler)
medyan = statistics.median(sureler)
mod = statistics.mode(sureler)
std = statistics.stdev(sureler)
q1, q2, q3 = statistics.quantiles(
    sureler, n=4, method="inclusive")
print(f"Ortalama: {ortalama:.2f} gün")
```

OUTPUT

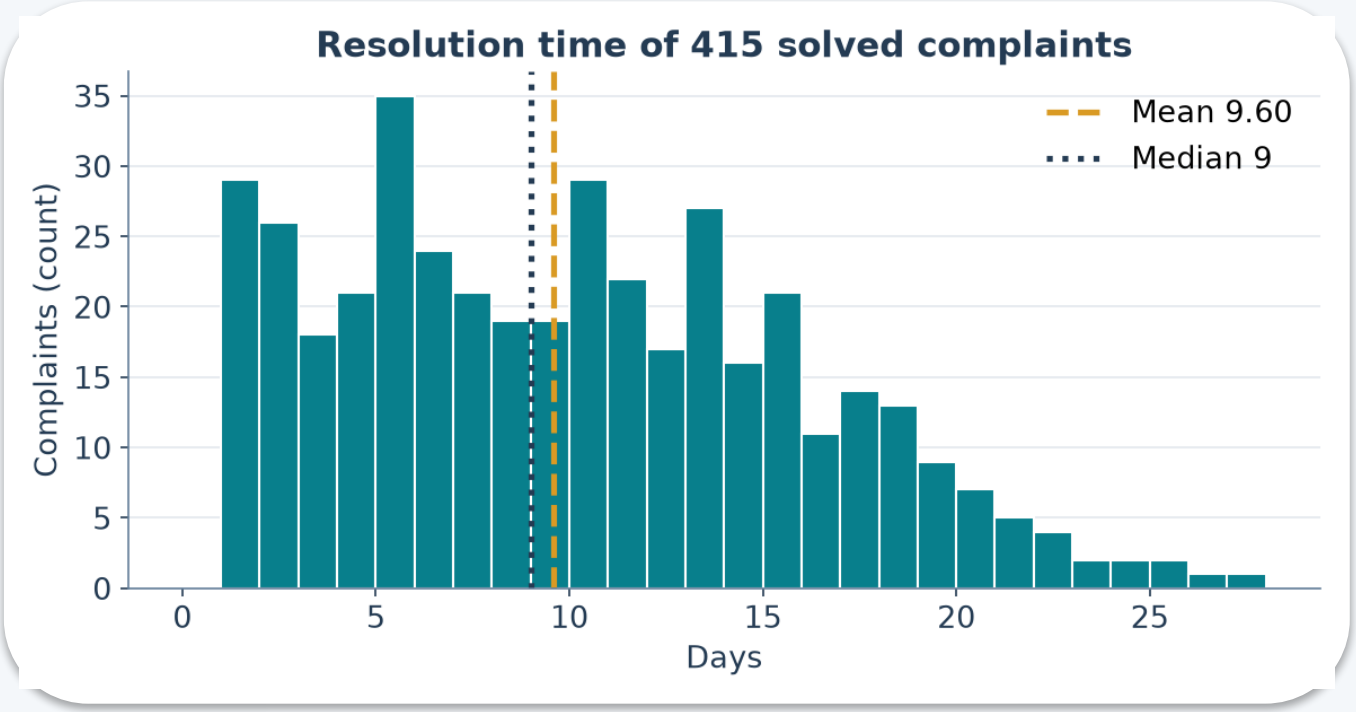
```
Ortalama: 9.60 gün
```

Official docs: docs.python.org/3/library/statistics.html

gun05/07_analiz_dusuncesi.py

Mean or median?

- Mean uses every value; long tails pull it up
- Median is the middle value
- Here: mean 9.60 > median 9 → right skew
- Report both, plus the count (n = 415)



Official docs: docs.python.org/3/library/statistics.html

gun05/08_hiz_standart.py

Many values at once: list vs NumPy

- List: a loop or comprehension for each value
- NumPy: one operation on the whole column
- A comparison gives a True/False mask
- Combine masks with & and brackets, not and

```
# standard library (08)
dusuk_mu = [i < 0.8 * p
            for i, p in zip(indirme, paket)]
print(sum(dusuk_mu))                    # 278

# NumPy (09)
import numpy as np
maske = indirme < 0.8 * paket           # arrays
etiket = np.where(maske, "DÜŞÜK", "UYGUN")
```

Official docs: numpy.org/doc/stable/user/absolute_beginners.html

gun05/10_pandas_ilk_bakis_opsiyonel.py

pandas: a DataFrame is an Excel sheet

#	sikayet_no	tarih	il	kategori	durum	cozum_suresi_gun
0	10001	2026-01-01	İstanbul	Fatura	Çözüldü	6.0
1	10002	2026-01-01	Ankara	Kesinti	İşlemde	NaN
2	10003	2026-01-02	İzmir	Kapsama	Çözüldü	18.0

- Columns have names and types
- The index is the row number
- Operations work on whole columns
- Look first: shape, head(), info(), describe()

```
import pandas as pd
df = pd.read_csv(ROOT / "data" / "sikayetler.csv")
print("Satır, sütun:", df.shape)      # (600, 11)
print(df.head())
df.info()
```

Official docs: pandas.pydata.org/docs/getting_started/intro_tutorials/index.html

gun05/10_pandas_ilk_bakis_opsiyonel.py

Filter, sort, new column

- df[condition] = AutoFilter
- Each condition in brackets, joined with &
- .isin([...]) = several ticked boxes
- A new column fills every row (drag down)

```
ist_kesinti = df[(df["il"] == "İstanbul")
                 & (df["kategori"] == "Kesinti")]
print("İstanbul + Kesinti:", len(ist_kesinti))
buyuk = df[df["il"].isin(["İstanbul", "Ankara"])]
en_uzun = df.sort_values("cozum_suresi_gun",
                         ascending=False).head(5)
df["uzun_mu"] = df["cozum_suresi_gun"] > 14
print(df["kanal"].value_counts())
```

Official docs: pandas.pydata.org/docs/getting_started/comparison/comparison_with_spreadsheets.html

gun05/11_pandas_gruplama_opsiyonel.py

groupby + agg = PivotTable

- groupby: rows of the pivot
- agg: values and their summary
- size counts rows; count skips NaN
- round(2) for a readable table

```
ozet = df.groupby("operator").agg(
    sikayet=("sikayet_no", "size"),
    ort_sure=("cozum_suresi_gun", "mean"),
    medyan_sure=("cozum_suresi_gun", "median"),
    ort_memnuniyet=("memnuniyet", "mean"),
).round(2)
print(ozet)
```

Official docs: pandas.pydata.org/docs/user_guide/groupby.html

gun05/11_pandas_gruplama_opsiyonel.py

pivot_table and merge (DÜŞEYARA)

- pivot_table: city × category, margins = Grand Total
- merge brings the target day to each row
- validate="many_to_one" checks the lookup table
- Row count after merge must stay 600

```
pivot = pd.pivot_table(df, index="il",
    columns="kategori", values="sikayet_no",
    aggfunc="count", fill_value=0, margins=True)
hedefler = pd.read_excel(xlsx,
                         sheet_name="Hedefler")
birlesik = df.merge(hedefler, on="kategori",
    how="left", validate="many_to_one")
print(len(birlesik))                    # 600
```

Official docs: pandas.pydata.org/docs/user_guide/merging.html

gun05/12_standart_filtre_grup.py

No pandas? The same pivot with a dictionary

- defaultdict(list): a group per key
- Collect values, then summarise each group
- Same numbers as pandas: 39.8%
- B route works on every laptop

```
from collections import defaultdict
import statistics
gruplar = defaultdict(list)
for k in kayitlar:
    sure = k["cozum_suresi_gun"]
    if sure is not None:
        gruplar[k["operator"]].append(sure)
for op, sureler in sorted(gruplar.items()):
    print(op, round(statistics.mean(sureler), 2))
```

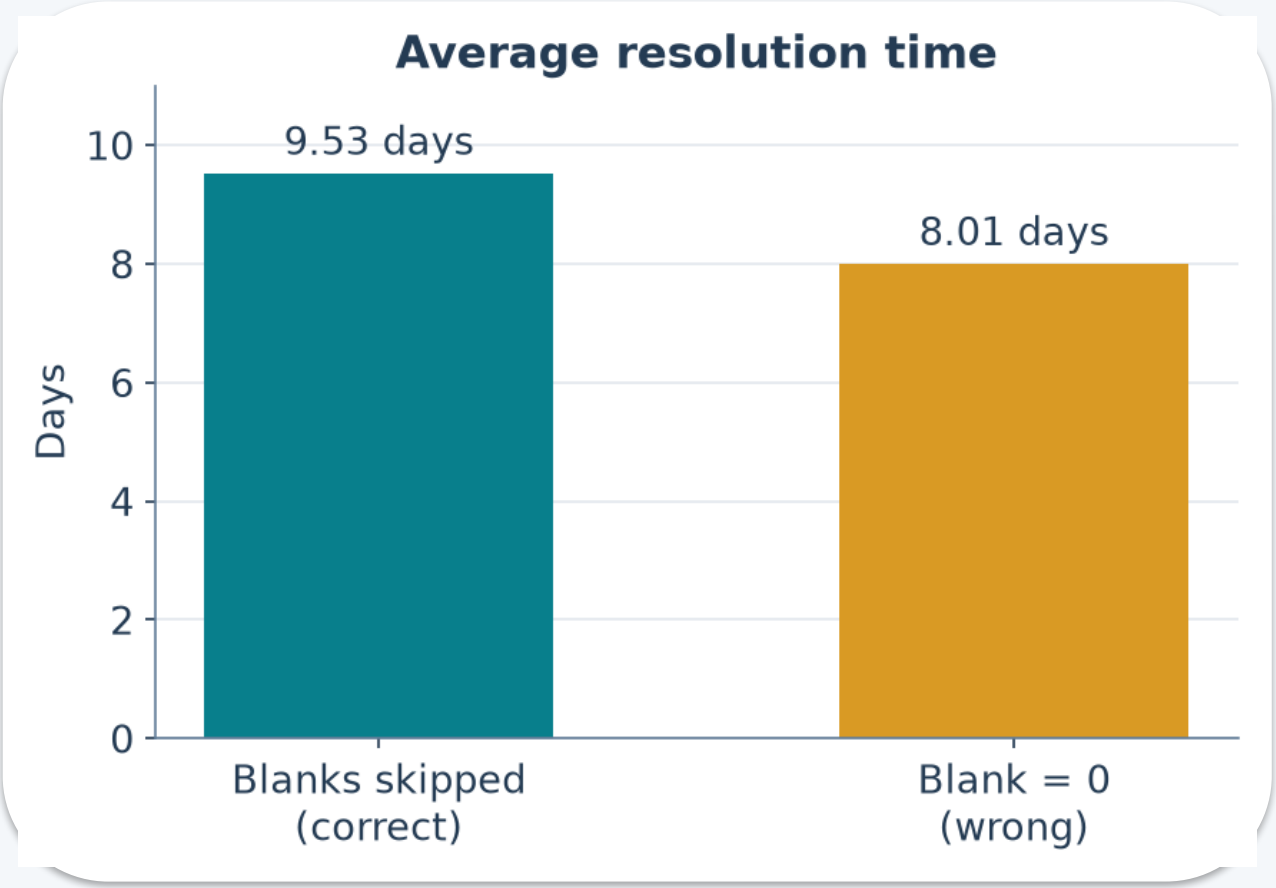
Official docs: docs.python.org/3/library/collections.html

gun05/13_eksik_veri_opsiyonel.py

Missing is not zero

- Blank time = not finished yet
- fillna(0) says 'finished in 0 days'
- Average drops from 9.53 to 8.01
- The business rule decides, not the code

```
sure = df["cozum_suresi_gun"]
dogru = sure.mean()             # 9.53
yanlis = sure.fillna(0).mean() # 8.01
```



Official docs: pandas.pydata.org/docs/user_guide/missing_data.html

gun05/15_kirli_veri_opsiyonel.py

Clean dirty data with rules

- strip spaces, fix spelling (title)
- to_numeric(errors="coerce") → NaN, not a crash
- Check ranges: memnuniyet 1-5
- Do not delete: write a 'to review' file

```
df["il"] = df["il"].str.strip().str.title()
df = df.drop_duplicates(subset="sikayet_no")
sure = pd.to_numeric(df["cozum_suresi_gun"],
                     errors="coerce")
kurallar = {
    "bilinmeyen il": ~df["il"].isin(gecerli_iller),
}
```

Official docs: pandas.pydata.org/docs/user_guide/missing_data.html

gun05/16_excel_oku_yaz_opsiyonel.py

Read and write Excel

- read_excel(sheet_name=None) → all sheets
- ExcelWriter → one workbook, many sheets
- No openpyxl? 17_xlsx_standart.py (B route)
- File open in Excel → PermissionError

```
sayfalar = pd.read_excel(kaynak, sheet_name=None)
dosya = cikti / "sikayet_raporu.xlsx"
with pd.ExcelWriter(dosya) as w:
    ozet.to_excel(w, sheet_name="Özet")
    pivot.to_excel(w, sheet_name="Pivot")
    detay.to_excel(w, sheet_name="Detay")
```

Official docs: pandas.pydata.org/docs/reference/api/pandas.ExcelWriter.html

Excel → Python (Day 5)

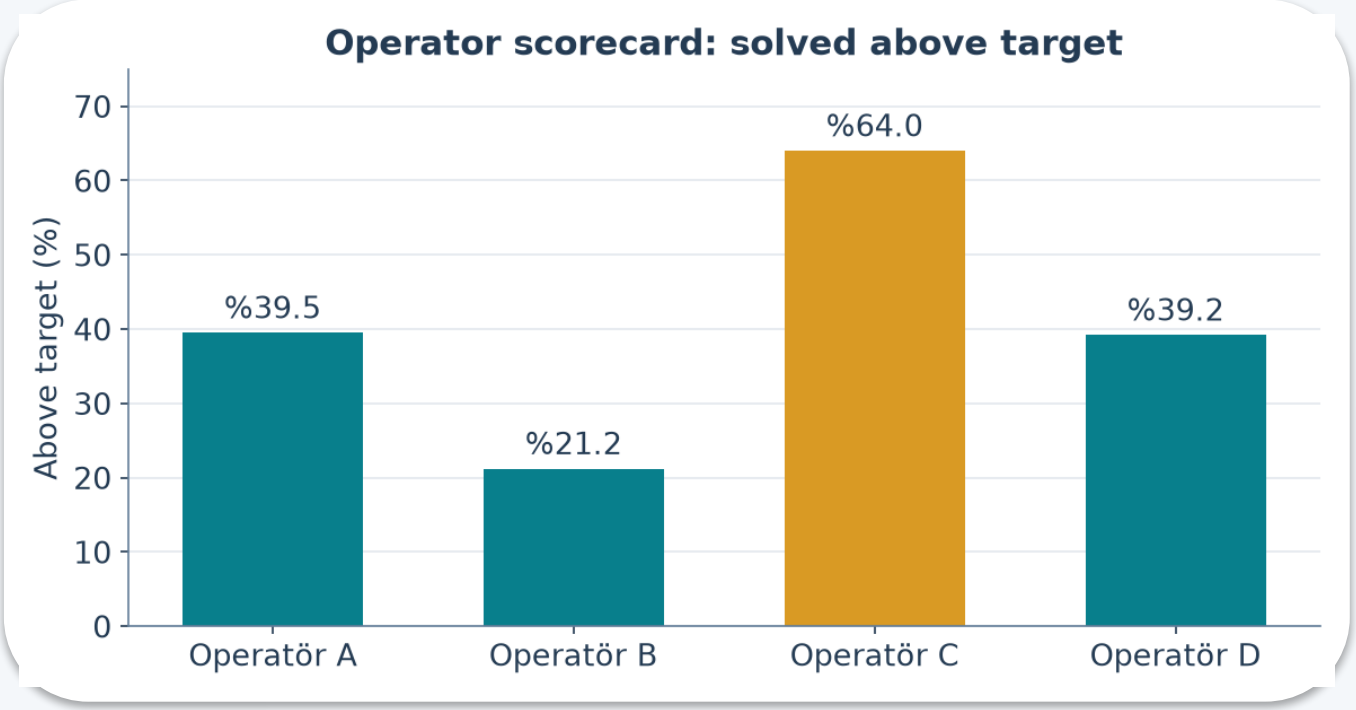
Excel	Standard library	pandas / NumPy
ORTALAMA / ORTANCA	`statistics.mean / median`	`.mean() / .median()`
STDSAPMA.S	`statistics.stdev`	`.std(), np.std(ddof=1)`
Filter	`if inside a comprehension`	`df[condition], query`
Sort	`sorted(key=...)`	`sort_values`
EĞERSAY	`Counter`	`value_counts`
PivotTable	`defaultdict + loop`	`groupby().agg(), pivot_table`
DÜŞEYARA	`hedef[kategori]`	`merge`
EĞER	`"A" if x else "B"`	`np.where`
Remove Duplicates	`set of seen IDs`	`drop_duplicates`

Official docs: pandas.pydata.org/docs/getting_started/comparison/comparison_with_spreadsheets.html

gun05/18_karne_bos.py

Mini project • operator scorecard

- Six measures per operator
- Solve rate = Çözüldü / all complaints
- Times and late % → solved only
- Is C 'the worst'? Check the category mix



Official docs: pandas.pydata.org/docs/user_guide/groupby.html

Exit ticket • Day 5

1

One finding: population, count, measure, unit

2

One limitation of that finding

3

Why not fillna(0) for resolution time?

4

How do you know two routes agree?

Official docs: docs.python.org/3/tutorial/index.html

DAY 06

Day 6 • Visualise and explain

1 Which question, which chart?

2 First good chart (live)

3 Sorted bars and time lines

4 Distribution and relationship

5 Dashboard, seaborn, pandas

6 Critique and exit ticket

TODAY'S GOAL

Question first, then the chart: title, axis labels, unit, and a bar axis that starts at 0.

`gun06/06_grafik_secimi.py`

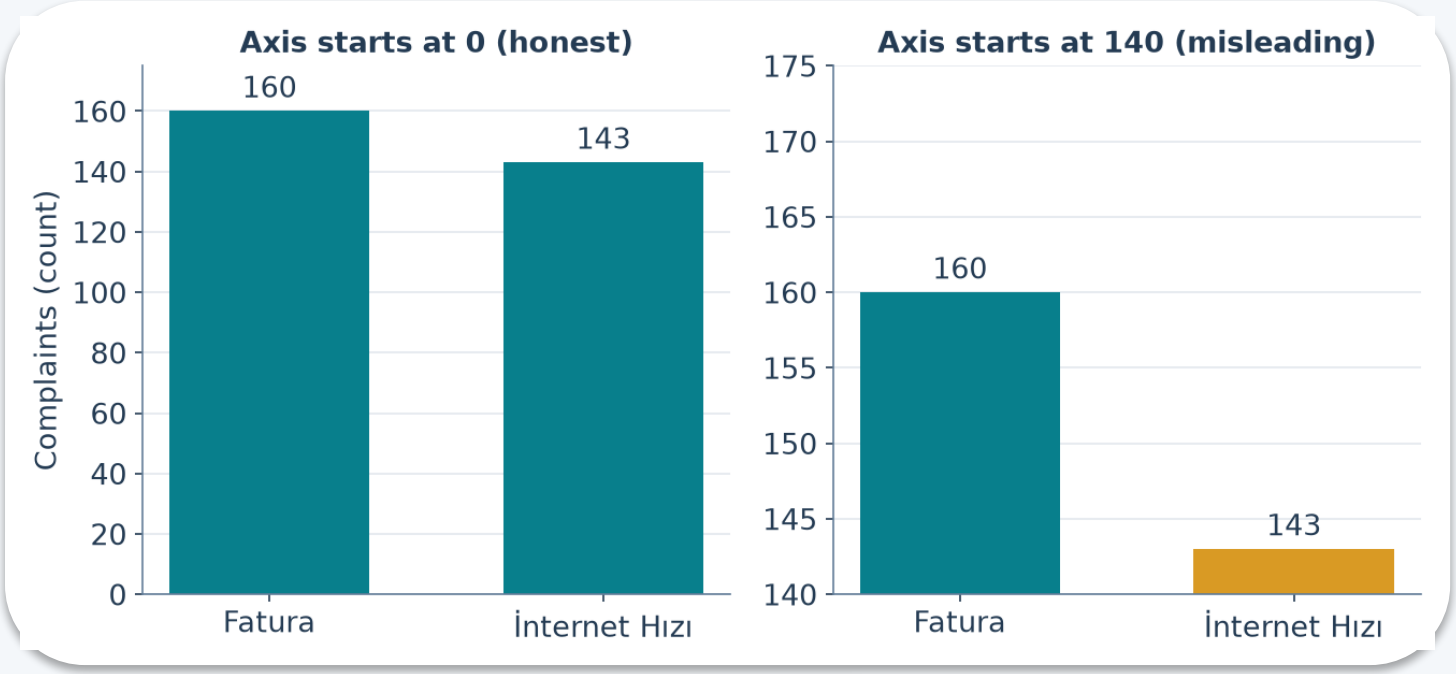
The question chooses the chart

Question	Chart	Files
Which is bigger? (compare)	bar / horizontal bar	07, 08, 18
How did it change? (time)	line	09, 17
How is it spread? (distribution)	histogram, box plot	10, 11
Does one rise with the other? (relationship)	scatter	12
What share? (part of whole)	stacked bar, (careful) pie	16, 06

Official docs: matplotlib.org/stable/users/explain/quick_start.html

`gun06/06_grafik_secimi.py`

A bar axis must start at 0

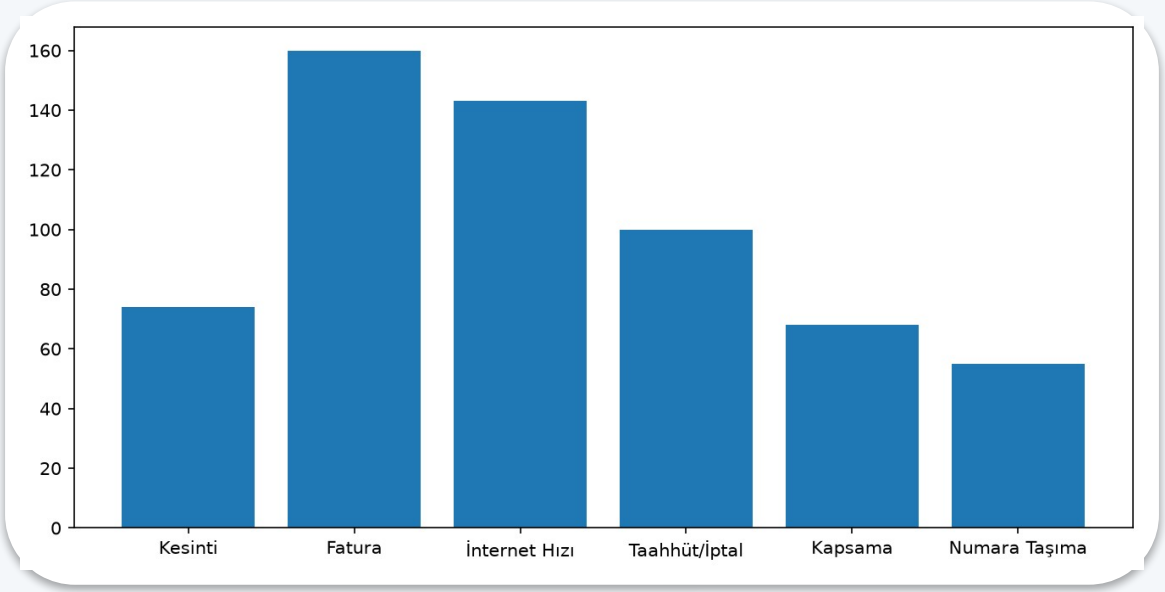


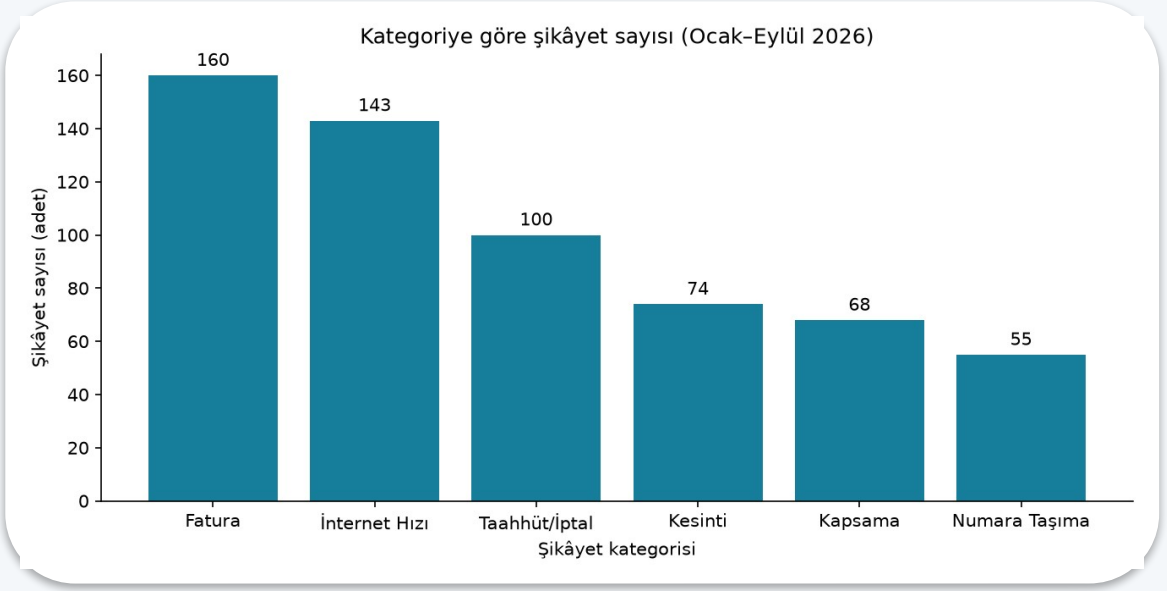
- Real ratio: 160 / 143 = 1.12
- From 140 the bars look 6.7 times apart
- Same data, a different story
- ax.set_ylim(bottom=0)

Official docs: matplotlib.org/stable/users/explain/quick_start.html

`gun06/07_ilk_grafik_bos_opsiyonel.py`

Live coding: from default to clear





`Start: 07_ilk_grafik_bos_opsiyonel.py`

`After TODO 1-6: 07_ilk_grafik_opsiyonel.py`

Same data, six small changes

Official docs: matplotlib.org/stable/users/explain/quick_start.html

gun06/07_ilk_grafik_opsiyonel.py

Matplotlib: figure, axes, save

- fig = the paper, ax = the drawing frame
- set_... sets title and labels
- bar_label writes the values
- savefig(dpi=150), then plt.close(fig)

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(9, 4.5))
cubuklar = ax.bar(kategoriler, adetler)
ax.set_title("Kategoriye göre şikâyet sayısı")
ax.set_ylabel("Şikâyet sayısı (adet)")
ax.set_ylim(bottom=0)
ax.bar_label(cubuklar, padding=3)
fig.savefig(cikti / "07_kategori_bar.png", dpi=150)
plt.close(fig)
```

Official docs: matplotlib.org/stable/api/_as_gen/matplotlib.figure.Figure.savefig.html

Checklist for every chart

- A title that states the question or the finding
- Axis labels with units: adet, gün, Mbps, %
- Bars start at 0
- One colour; a second only to highlight
- Turkish letters display correctly
- A source / limitation note: synthetic data, Jan–Sep 2026

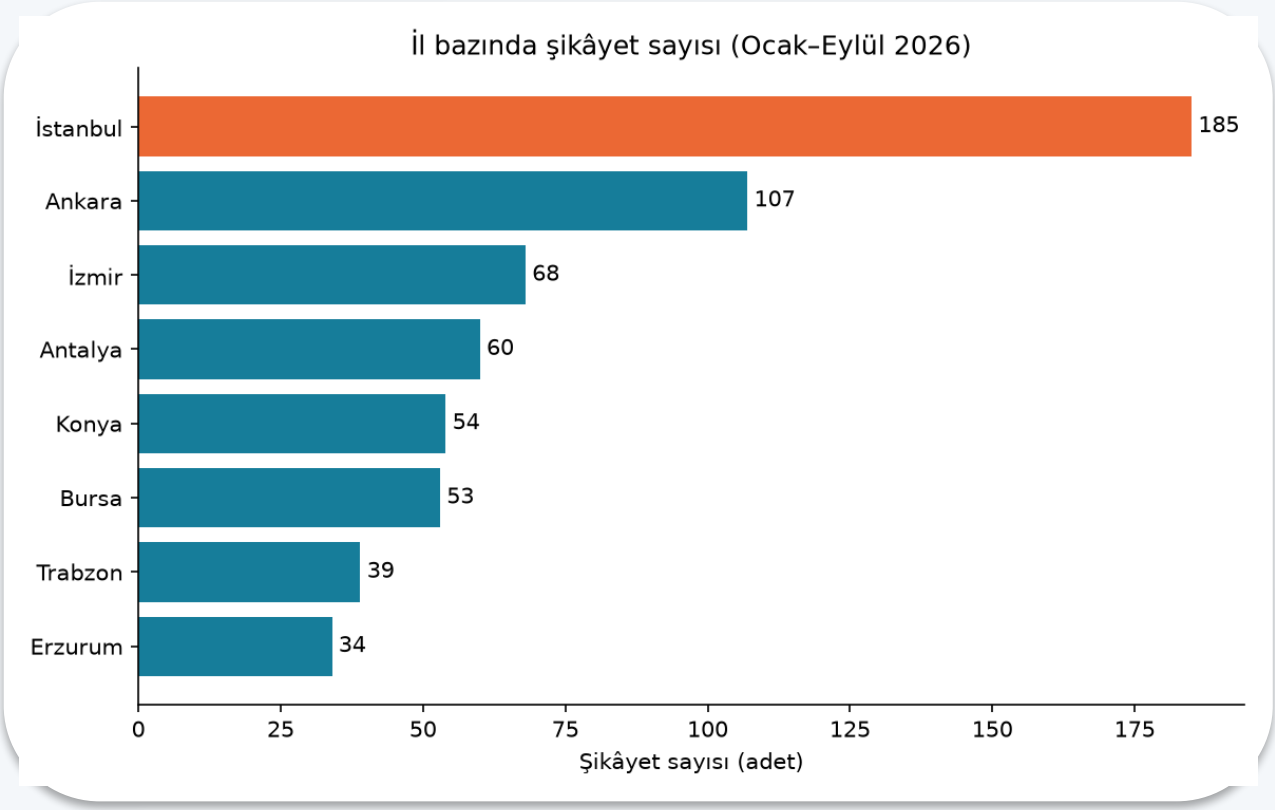
Official docs: matplotlib.org/stable/users/explain/quick_start.html

gun06/08_sirali_barh_opsiyonel.py

Sorted bars and one highlight

- barh draws from the bottom up → sort ascending
- Highlight one bar; the rest stay neutral
- Most complaints ≠ most problematic: İstanbul is the biggest city

```
sirali = sorted(il_sayilari.items(),
                key=lambda t: t[1])
renkler = ["#eb6834" if il == en_yogun
           else "#167d9a" for il in iller]
ax.barh(iller, adetler, color=renkler)
```

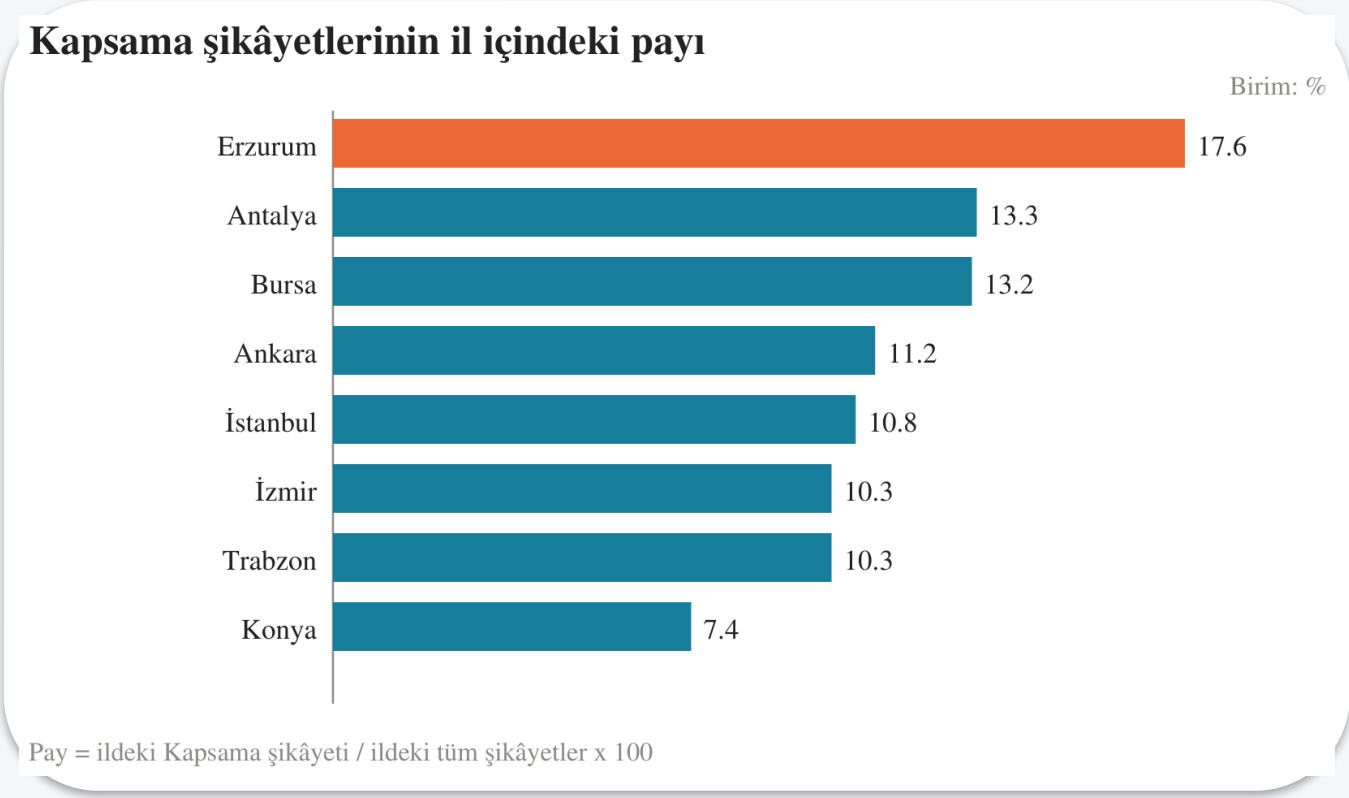


Official docs: matplotlib.org/stable/api/_as_gen/matplotlib.axes.Axes.bar_label.html

`gun06/18_svg_yatay_bar.py`

Count or share? Ask the right question

- Count: İstanbul first (biggest city)
- Share of Kapsama inside each city: Erzurum first
- Share = city's Kapsama / city's all × 100
- Small groups → shares jump around



Official docs: developer.mozilla.org/en-US/docs/Web/SVG/Tutorial

gun06/09_cizgi_trend_opsiyonel.py

Line chart: keep time in order

- Month from the date text: tarih[:7] = SOLDAN(A2;7)
- Sort the months before plotting
- Two series need a legend
- annotate: point at the peak

```
toplam = Counter(k["tarih"][:7]
                 for k in kayitlar)
aylar = sorted(toplam)
ax.plot(etiketler, toplam_degerler,
        marker="o", label="Tüm şikâyetler")
ax.legend()
```



Official docs: matplotlib.org/stable/api/_as_gen/matplotlib.axes.Axes.plot.html

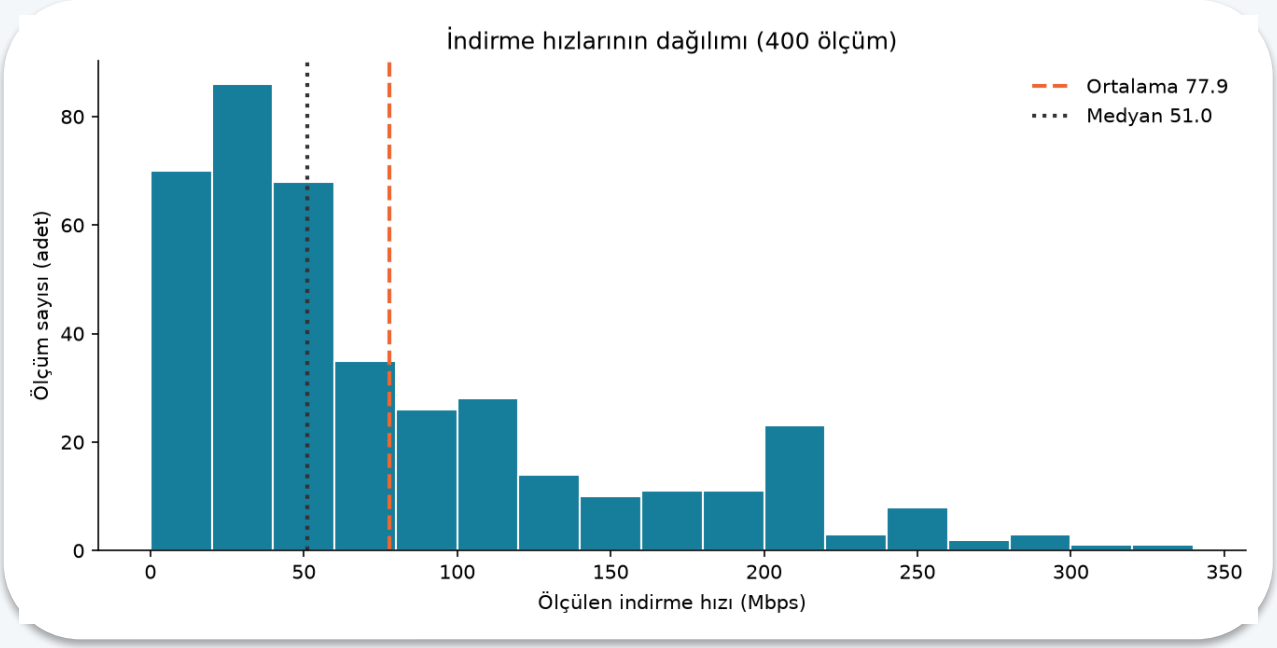
gun06/10_histogram_opsiyonel.py

Histogram: how values spread

- x axis = number ranges (bins), not categories

- Bar height = how many values in that range
- Mean 77.9 > median 51.0 → a long right tail
- Bin width changes the picture

```
sinirlar = list(range(0, 341, 20))
ax.hist(hizlar, bins=sinirlar)
ax.axvline(medyan, linestyle=":",
           label="Medyan")
```

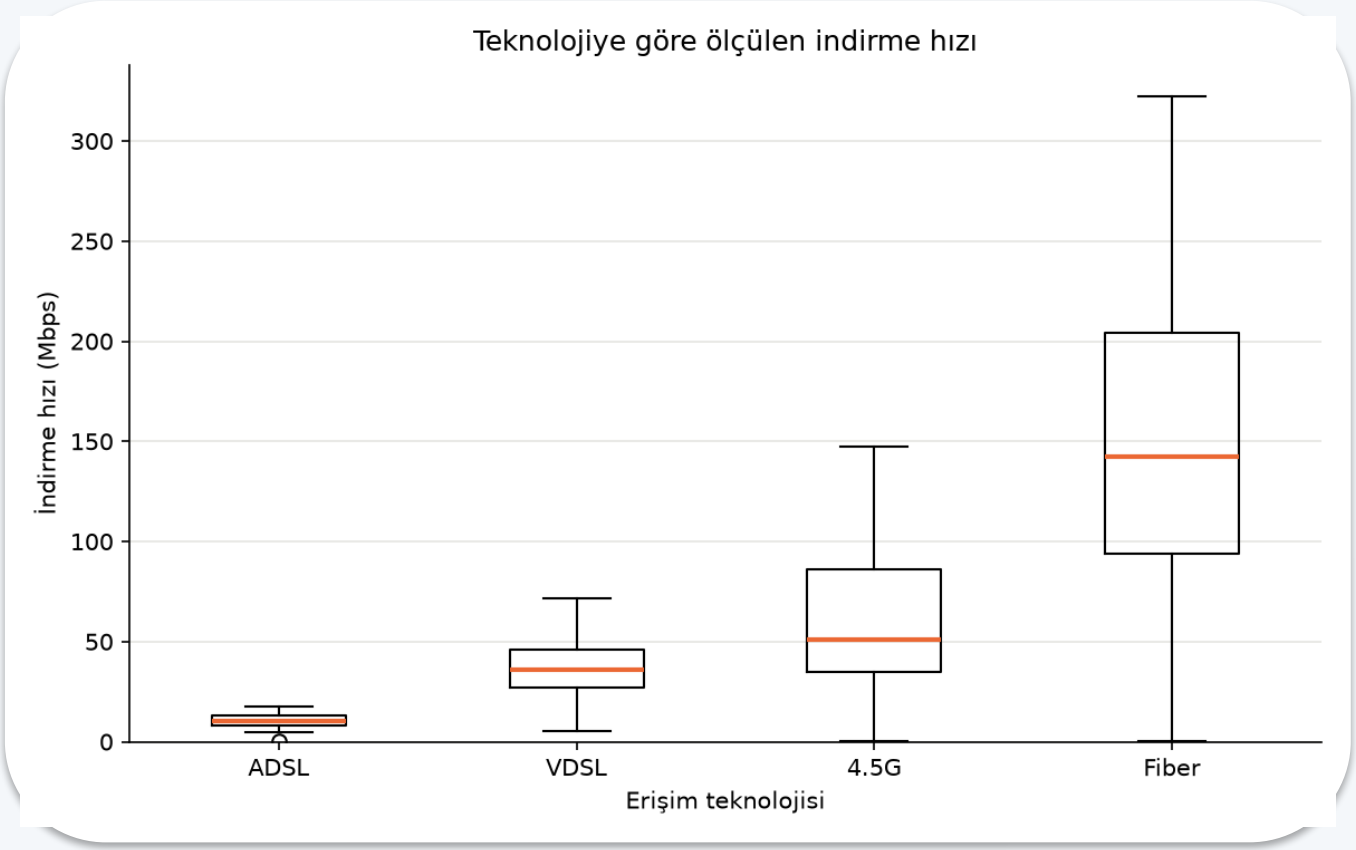


Official docs: matplotlib.org/stable/api/_as_gen/matplotlib.axes.Axes.hist.html

`gun06/11_boxplot_opsiyonel.py`

Box plot: compare spreads

- Line in the box = median
- Box = middle 50% (Q1 to Q3)
- Whiskers = typical range; dots = outliers
- Fiber: highest median, widest spread

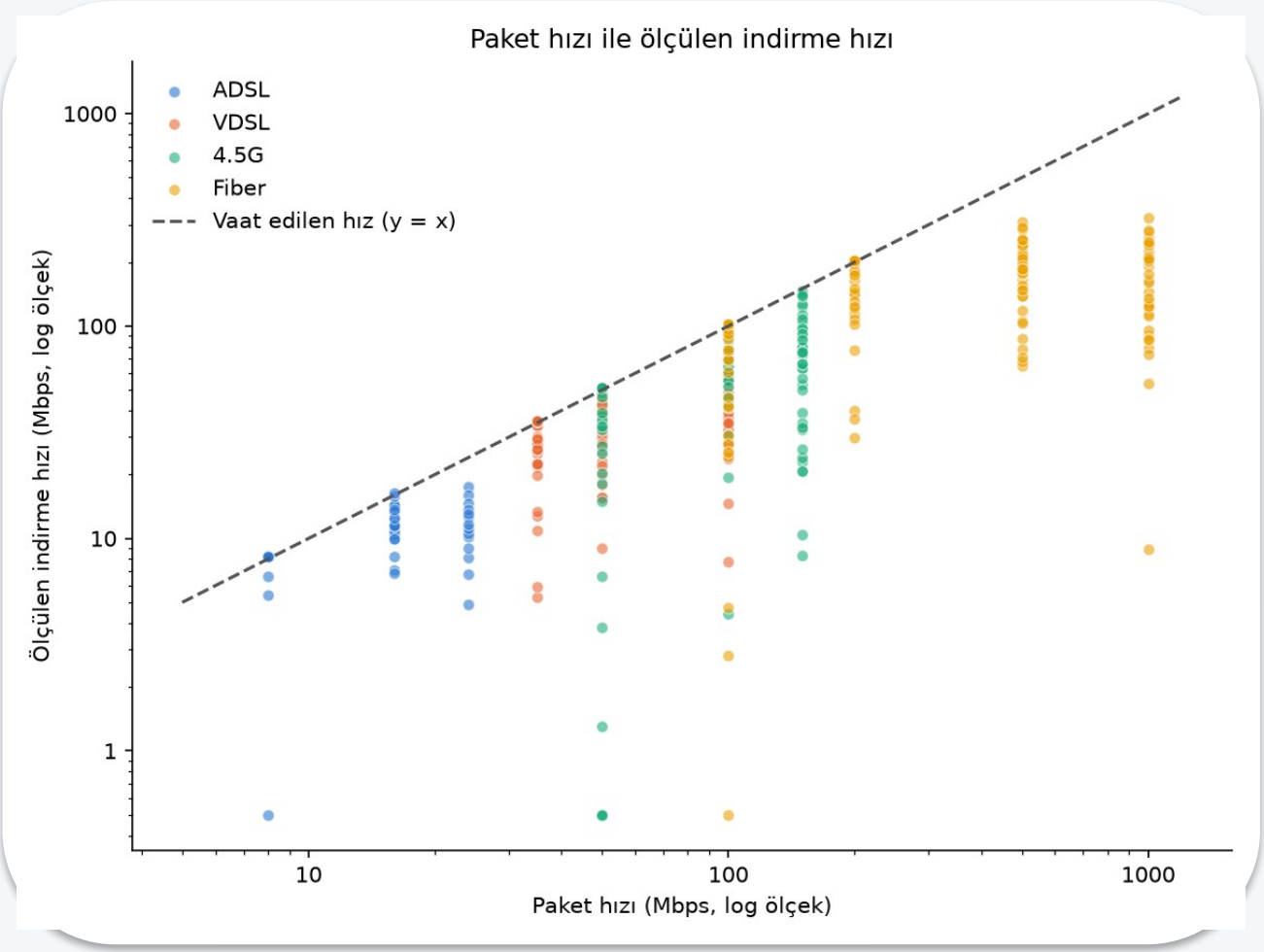


Official docs: matplotlib.org/stable/api/_as_gen/matplotlib.axes.Axes.boxplot.html

`gun06/12_scatter_opsiyonel.py`

Scatter: is there a relationship?

- Each dot = one measurement
- Dashed line y = x: the promised speed
- Below the line = less than promised
- Log axes: each grid line ×10



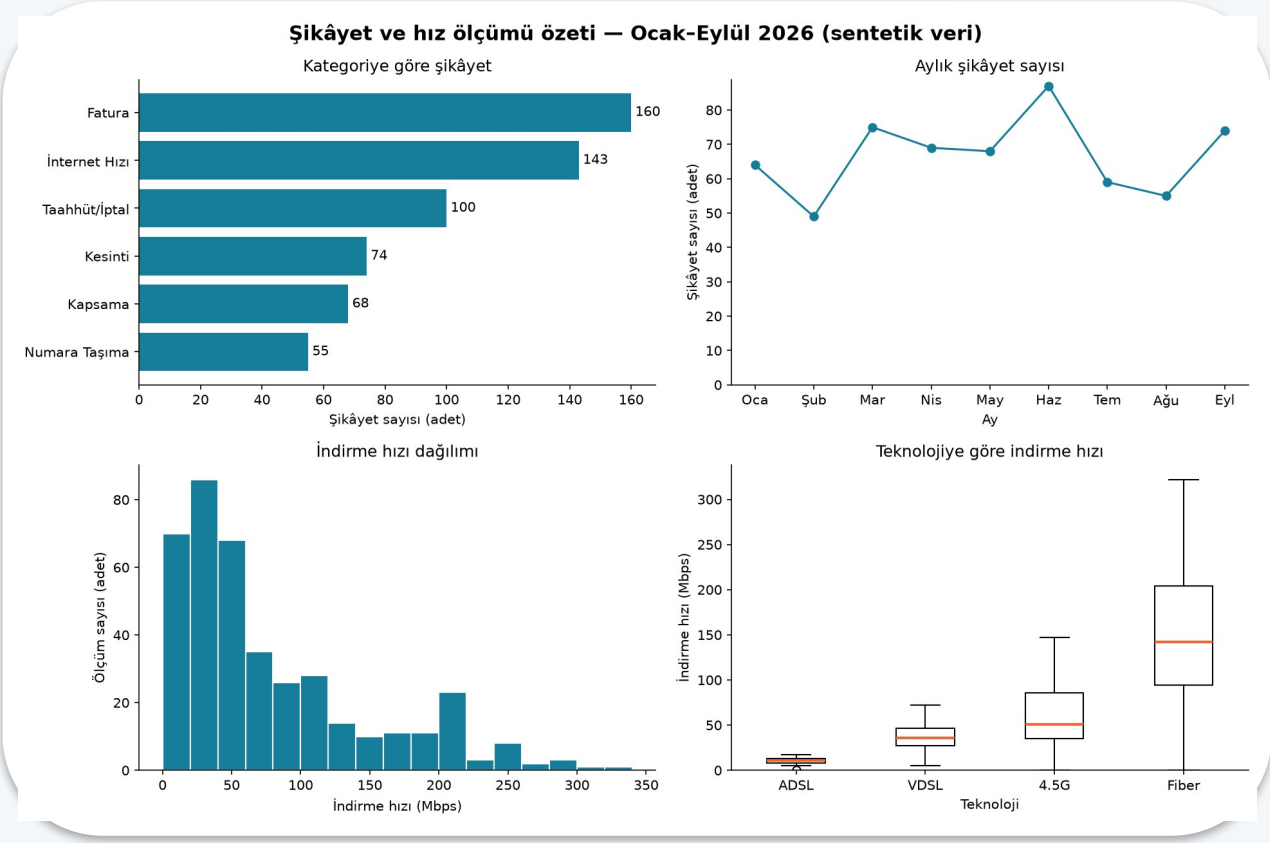
Official docs: matplotlib.org/stable/api/_as_gen/matplotlib.axes.Axes.scatter.html

gun06/13_dashboard_opsiyonel.py

One page, four charts

- plt.subplots(2, 2) → a grid of axes
- axler[0, 1] = row 0, column 1 (like a cell)
- for ax in axler.flat → same style for all
- fig.suptitle: one title for the page

```
fig, axler = plt.subplots(2, 2)
ax = axler[0, 0]
ax.barh(kategori_adlari, adetler)
```



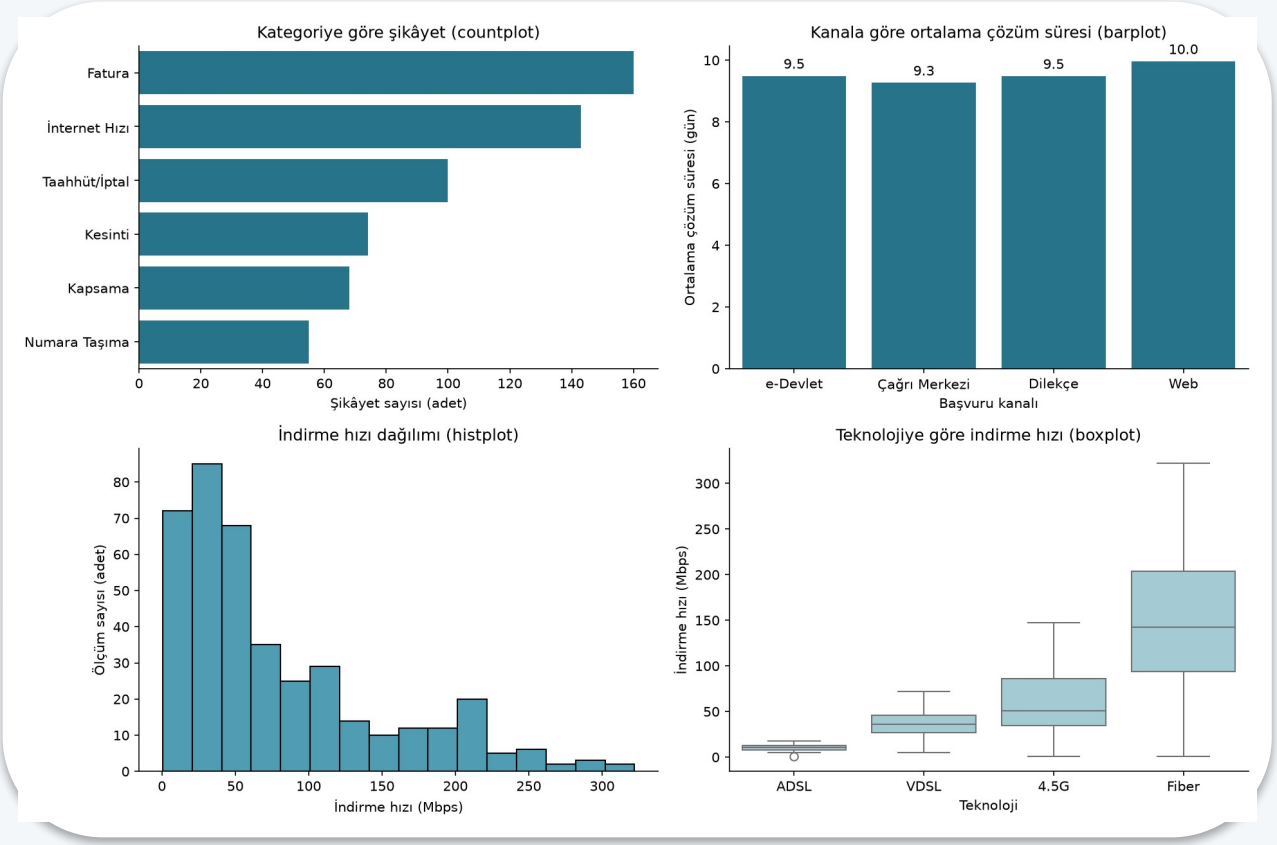
Official docs: matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.subplots.html

gun06/14_seaborn_temel_opsiyonel.py

seaborn: name the columns, it counts

- data=DataFrame, x= / y= column names
- countplot counts, barplot averages
- errorbar=None hides the 95% interval
- order= keeps the bars sorted

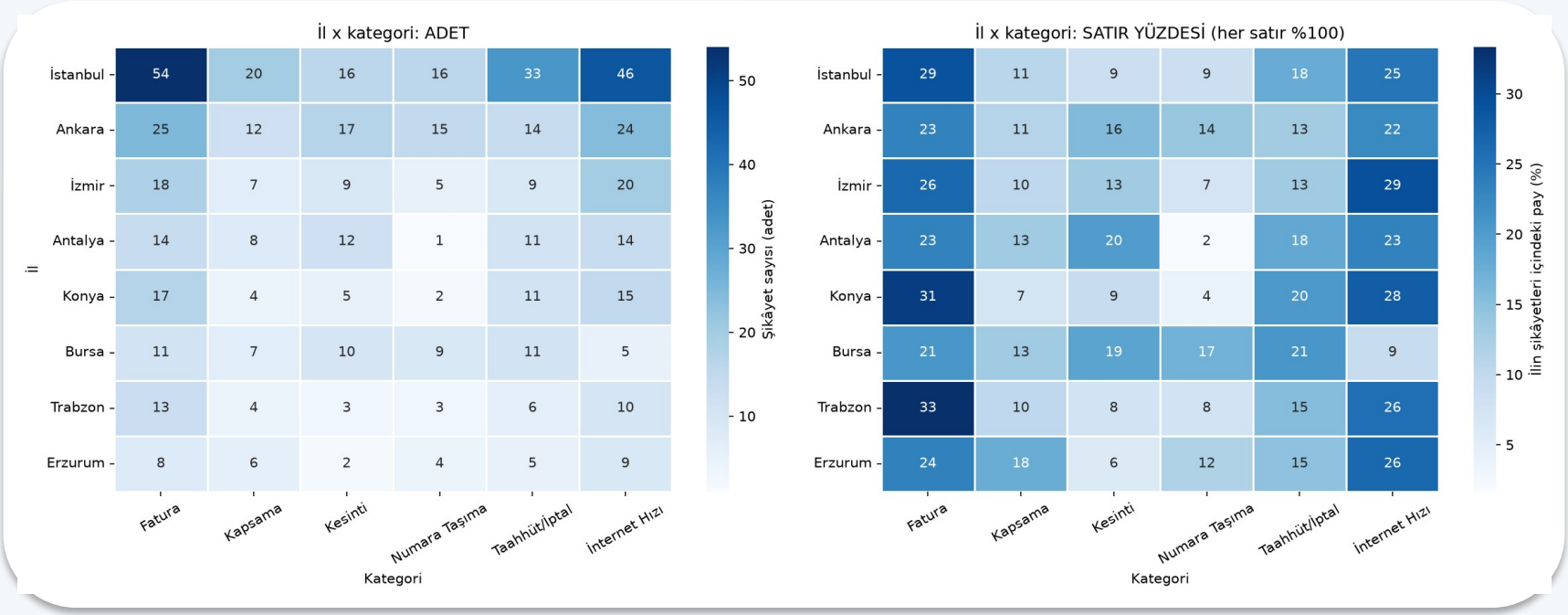
```
import seaborn as sns
sns.countplot(data=sikayet,
              y="kategori",
              order=kategori_sirasi,
              ax=axler[0, 0])
```



Official docs: seaborn.pydata.org/tutorial.html

`gun06/15_seaborn_heatmap_opsiyonel.py`

Heatmap: count vs row share

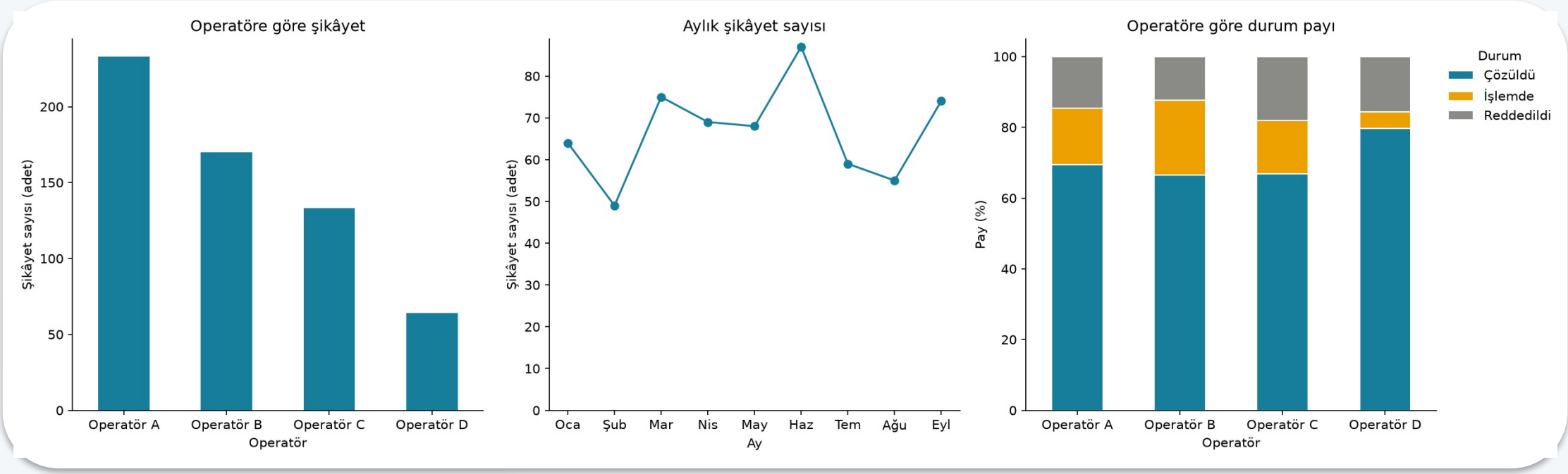


- pivot_table: city × category
- Left: counts (big cities dominate)
- Right: each row = 100%
- Like Power BI matrix + conditional format

Official docs: seaborn.pydata.org/generated/seaborn.heatmap.html

`gun06/16_pandas_plot_opsiyonel.py`

pandas .plot() for quick looks



- df.plot(kind="bar") — one line, fast exploration
- For reports, finish the labels by hand
- 100% stacked bar answers a share question

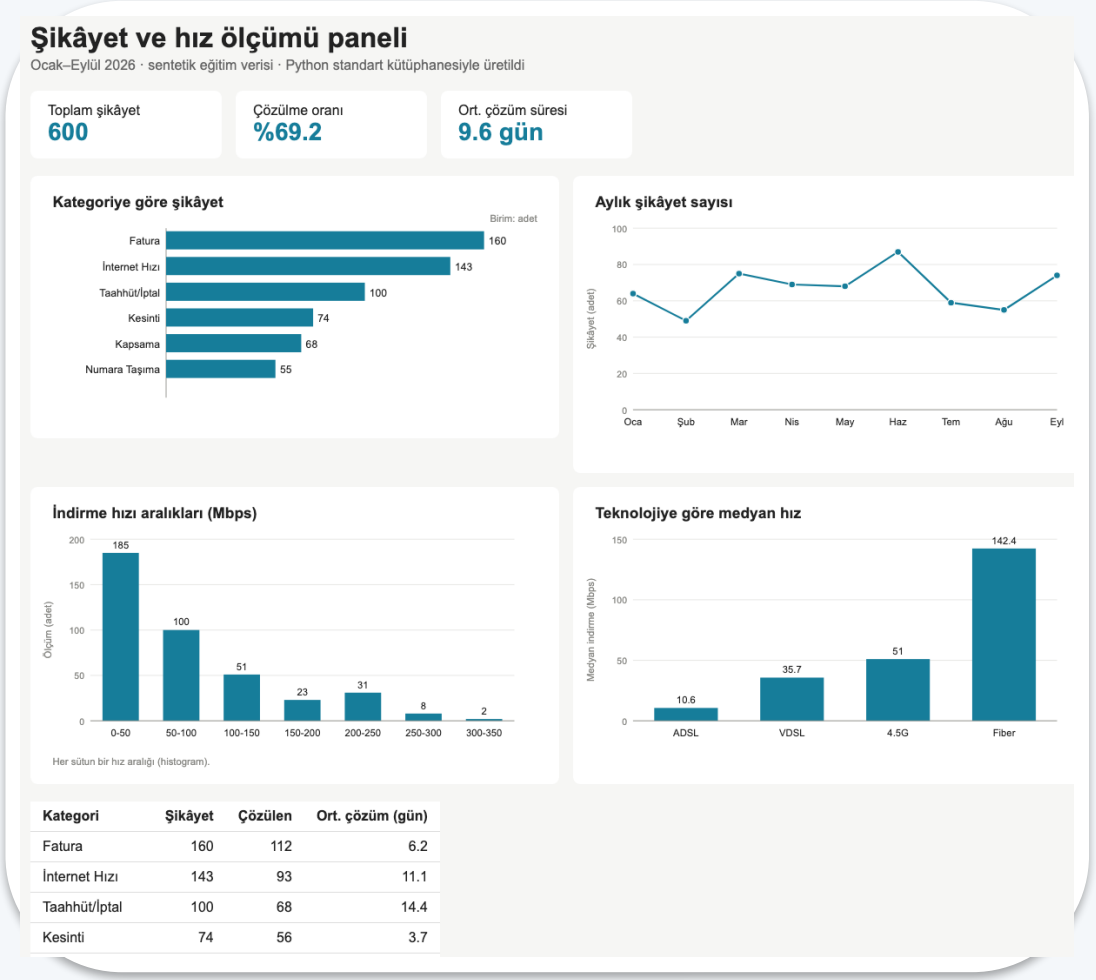
Official docs: pandas.pydata.org/docs/user_guide/visualization.html

gun06/19_html_dashboard.py

No packages? SVG and HTML with plain Python

- svg_cizim.py: ready drawing functions
- Call them like an Excel function: arguments in, file out
- SVG opens in Edge; stays sharp at any size
- 19 → one HTML dashboard with KPI boxes

```
from svg_cizim import kaydet, yatay_bar_svg
svg = yatay_bar_svg(
    iller, adetler,
    baslik="İl bazında şikâyet sayısı",
    birim="adet")
kaydet(svg, "18_il_bar.svg")
```



Official docs: developer.mozilla.org/en-US/docs/Web/SVG/Tutorial

Excel chart → Python call

Excel chart	matplotlib / seaborn
Column chart	`ax.bar(x, y)`
Bar chart	`ax.barh(y, x)`
Line chart	`ax.plot(x, y, marker="o")`
Histogram	`ax.hist(values, bins=...)`
Box and whisker	`ax.boxplot(groups)`
Scatter (XY)	`ax.scatter(x, y)`
Conditional-format matrix	`sns.heatmap(pivot, annot=True)`
Save as picture	`fig.savefig("x.png", dpi=150)`

Official docs: matplotlib.org/stable/users/explain/quick_start.html

Practice • one chart, one finding

CORE T1: complaints per operator, checklist complete

CORE T3: critique a chart (no code)

MIDDLE O1: monthly line of solved complaints only

EXTRA İ1-İ3 in ALISTIRMALAR.md

```
py gun06/cozumler.py
py gun06/cozumler_ek.py
```

OUTPUT

```
Gün 6 ek çözümler tamamlandı.
```

Official docs: matplotlib.org/stable/users/explain/quick_start.html

Exit ticket • Day 6

1

Which chart for a question about time?

2

Why does a bar axis start at 0?

3

One finding from your chart

4

One limitation of that chart

Official docs: docs.python.org/3/tutorial/index.html

DAY 07

Day 7 • A complete local workflow

1 Read and adapt a ready script

2 Parameters with argparse

3 End-to-end pipeline

4 Team project: choose a card

5 Team work and checking

6 Presentations and close

TODAY'S GOAL

Write the flow once; every month one command gives the same report — checked two ways.

`gun07/07_pipeline_stdlib.py`

The final workflow



- Keep the input file unchanged
- Each step is a small function
- Run again tomorrow → the same result

Official docs: docs.python.org/3/library/sqlite3.html

gun07/05_hazir_script_uyarlama.py

Adapt a ready-made script

- Read the first comments
- Change only the SETTINGS block
- Run, compare, then change one more thing
- Like the constants at the top of an Excel macro

```
# ---- AYARLAR (only change this part) ----
IL = "Ankara"
KATEGORI = "İnternet Hızı"
ESIK_GUN = 10
CIKTI_ADI = "05_esik_asanlar.csv"
# -----------------------------------------
```

OUTPUT

```
Ankara / İnternet Hızı: 24 şikâyet, 17 çözülen
```

Official docs: docs.python.org/3/tutorial/index.html

`gun07/06_filtre_parametreleri.py`

Settings in the file or on the command line?

05: edit the file

- Open the .py and change IL
- Good for your own copy
- Easy to forget what you changed

06 / 07: give parameters

- py ... --il Ankara
- Nobody edits the file
- --help explains the options

Official docs: docs.python.org/3/howto/argparse.html

gun07/06_filtre_parametreleri_bos.py

argparse: named parameters with help

- --il, --operator, --baslangic, --bitis
- default=None → filter not applied
- type=date.fromisoformat checks the date
- --help is written for you

```
import argparse
parser = argparse.ArgumentParser(
    description="Şikâyetleri süzer.")
parser.add_argument("--il", default=None)
parser.add_argument("--operator", default=None)
parser.add_argument("--baslangic",
                    type=date.fromisoformat)
args = parser.parse_args()
```

Official docs: docs.python.org/3/howto/argparse.html

gun07/07_pipeline_stdlib.py

One pipeline, many reports

- No arguments → all data
- Values with spaces need double quotes
- Dates as YYYY-MM-DD
- --girdi reads another file (CSV or xlsx)

```
py gun07/07_pipeline_stdlib.py
py gun07/07_pipeline_stdlib.py --il Ankara
   --baslangic 2026-04-01 --bitis 2026-06-30
py gun07/07_pipeline_stdlib.py
   --operator "Operatör B"
py gun07/07_pipeline_stdlib.py
   --girdi data/sikayetler_kirli.csv
```

Official docs: docs.python.org/3/howto/argparse.html

`gun07/07_pipeline_stdlib.py`

One command → HTML report + Excel summary

- rapor_stdlib.html: numbers, tables, SVG charts
- ozet_stdlib.xlsx: summary sheets for Excel / Power BI
- pipeline_stdlib.sqlite: the clean data
- Hand-made: one hour • scripted: one second



Şikâyet analiz raporu

Kaynak: sikayetler.csv · Filtre: yok (tüm veri) · Sürüm: standart kütüphane (B planı) · Sentetik veri

Ana sayılar

Kayıt sayısı	600
Çözülen	415
Reddedilen	89
İşlemde	96
Çözülme oranı (%)	69.2
Ort. çözüm süresi (gün)	9.6
Medyan çözüm süresi (gün)	9.0
Hedef süreyi aşan çözüm (%)	39.8
En çok şikâyet kategorisi	Fatura
En yoğun il	İstanbul
En hızlı kanal	Çağrı Merkezi
Sorunlu (ayrılan) kayıt	0

Grafikler

Kategoriye göre şikâyet

Birim: adet

Official docs: docs.python.org/3/library/html.html

gun07/07_pipeline_stdlib.py

Inside the pipeline: one function per step

- Each step can be tested alone
- Errors point to one small function
- Day 4 SQL, Day 5 rules, Day 6 charts — reused

```
def main():
    args = parametreleri_al()
    ham = oku(girdi)                          # 1
    temiz, sorunlu = temizle(ham)             # 2
    secilen = filtrele(temiz, args)           # 3
    baglanti = sqlite_yaz(secilen, hedefler,
                          db_yolu)            # 4
    genel, kategori, aylik, kanal, il = \
        sorgula(baglanti)                     # 5
```

Official docs: docs.python.org/3/library/sqlite3.html

`gun07/07_pipeline_stdlib.py`

Check the main numbers

Measure	All data	Dirty file
Rows read / clean	600 / 600	26 / 20
Problem rows	0	6 (reasons listed)
Solved / Rejected / In progress	415 / 89 / 96	13 / 3 / 4
Solve rate	69.2%	65.0%
Mean / median days (solved)	9.6 / 9.0	9.46 / 10.0
Above target (solved)	39.8%	46.2%

A run is correct only when the numbers are checked

Official docs: docs.python.org/3/library/statistics.html

`gun07/09_iki_yol_karsilastir_opsiyonel.py`

Two routes must agree

07 standard library

- csv + sqlite3 + statistics
- SVG charts
- basit_xlsx.py for Excel
- Runs on every laptop

08 pandas (optional)

- read_excel, to_sql, read_sql
- matplotlib PNG in the HTML
- ExcelWriter
- Shorter code

Official docs: pandas.pydata.org/docs/getting_started/intro_tutorials/index.html

Team project • choose one card

Card	Business question	Data
1	Where are coverage (Kapsama) complaints concentrated?	sikayetler.csv: il, kategori
2	Do fiber packages deliver the promised speed?	hiz_olcumleri.csv
3	Which channel gets a faster resolution?	sikayetler.csv: kanal, süre
4	Which operator misses target days most?	sikayetler.csv + Hedefler sheet

Official docs: docs.python.org/3/tutorial/index.html

How the project is assessed

Criterion	0	2	4
Question and definition	missing	unclear measure	clear question + definition
Right data and filter	wrong subset	right, no check	right + one number checked by hand
Result correctness	contradicts	small error	consistent
Chart quality	no chart	labels / unit missing	right type, labelled, starts at 0
Interpretation	numbers only	finding, no limit	2 findings + 1 limitation

Bonus +2: parameters (argparse), meaningful file names, another team can run it.

Official docs: docs.python.org/3/tutorial/index.html

A five-minute team presentation

1. Question and definition (30 s)
2. Method: data, filter, steps (1 min)
3. Chart and two findings (2 min)
4. One limitation (30 s)
5. Questions (1 min)

Official docs: docs.python.org/3/tutorial/index.html

`gun07/04_streamlit_opsiyonel.py`

Optional • a local Streamlit app

What it is

- A browser page on your own laptop
- Python runs locally (localhost)
- Widgets: choose a file, a threshold

What to know

- A third-party package: needs approval
- A widget change reruns the script
- Start read-only; the CLI report is enough

Official docs: docs.streamlit.io/get-started/fundamentals/main-concepts

Seven days, one map

1. Day 1-2: types, decisions, loops, dictionaries
2. Day 3-4: functions, files, SQL
3. Day 5-6: clean, analyse, chart
4. Day 7: one command, one checked report

Exit ticket: the first report I will automate at work is _____

Official docs: docs.python.org/3/tutorial/index.html

Keep a reliable habit

- Keep one working copy
- Change one thing at a time
- Check numbers before conclusions
- Read the official documentation linked on each slide
- Thank you!

Official docs: docs.python.org/3/tutorial/index.html